

WDW-1

Connection-Typed Canonical Projection of the Frozen Einstein–Source Realization

ADM constraint architecture from the AASC physical interior

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Exact scope. The theorem constructs a source-faithful ADM prequotient of the accepted AASC Einstein–source realization on the declared regular, globally hyperbolic target. It establishes the canonical variables, constraints, multiplier structure, and local reconstruction interface. It does not claim refoliation closure, a global reduced phase space, quantization, or a Wheeler–DeWitt operator.

Abstract

This paper constructs the canonical projection of the accepted AASC Einstein–source realization. The dependency order is fixed upstream: non-degenerate determinate construction instantiates the AASC kernel; the realized physical interior supplies the regular constraint-governance role; standing-preserving finite transport yields connection type on the standard finite linear branch; and, on the explicitly certified minimal local diffeomorphism–natural first-jet metric subtype, the connection is Levi–Civita. Beginning only after those results, we freeze a smooth globally hyperbolic four-dimensional Lorentzian target and an attachment-certified, Cauchy-admissible Module-A source satisfying the exact Ward/divergence ledger. A $3 + 1$ decomposition then produces the induced spatial metric, extrinsic curvature, canonical momentum, lapse, shift, source projections, gravitational Hamiltonian constraint, and momentum constraint. We prove local reconstruction equivalence between the covariant Einstein–source equation and the resulting constraint-plus-evolution system, with all sign, regularity, topology, and boundary conventions declared.

The universal metric–source split is distinguished from a total matter-coupled Hamiltonian system. The latter is licensed only relative to a strong canonical source-family certificate specifying matter variables, the coupled-to-split symplectic comparison, Legendre map, a complete Dirac ledger, additive deformation constraints in the selected variables, and a map from complete canonical solution histories to the structured Module-A source quotient. Stress-tensor, Ward, boundary, and regularity matching are required history by history. For the minimally coupled Klein–Gordon model this realization map is constructed explicitly: each canonical solution is sent to its own attached structured source class, while the exact class used by the frozen Einstein–source realization is occupied by a named canonical history. No full phase space is identified with one fixed source fibre, and no class-wide matter lift is inferred from the example. On the compact boundaryless core target, lapse and shift are bulk multipliers and the foliation label is not thereby a physical clock. In boundary or asymptotic extensions, the corresponding data remain standing-bearing whenever they pair with nonzero differentiable generators or charges. The result is a connection-typed prequotient canonical reconstruction. It is not a primitive quantization of gravity, a complete field-theoretic reduced phase space, a proof of refoliation closure, or a Wheeler–DeWitt operator. It supplies a source-faithful typed family of classical handoffs whose source index sets form a disjoint exhaustive partition: the vacuum and strongly certified nonvacuum canonical branches are eligible for the later spatial-quotient and hypersurface-deformation papers, while an attached noncanonical nonvacuum source remains a reconstructive forced-history branch until a pointed strong lift is supplied.

Keywords: AASC; admissibility; canonical gravity; ADM decomposition; Einstein equation; Hamiltonian constraint; momentum constraint; source attachment; stress-energy; Levi–Civita connection; problem of time; Wheeler–DeWitt program.

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Read this first: exact endpoint

WDW-1 endpoint. On a frozen globally hyperbolic regular four-dimensional Lorentzian target in the explicitly certified minimal first-order natural metric subtype, the accepted Einstein equation with an attachment-certified smooth closed source admits a connection-typed ADM prequotient presentation. The normal and tangential projections give the geometric constraints, the Einstein–Hilbert representation gives the gravitational Legendre and multiplier structure, and the constraint-plus-evolution system reconstructs the full structured covariant Einstein-source object fibrewise. A total matter Hamiltonian exists only on the separately certified strong source-canonical-lift subtype; otherwise the source remains a covariantly prescribed forced-history object rather than a Hamiltonian constraint system.

Established in WDW-1	Not established in WDW-1
Levi–Civita ancestry on the declared minimal metric branch	A unique connection in torsion, metric-affine, Palatini, teleparallel, higher-jet, or nonlocal targets
Attachment-safe smooth Cauchy projections (ρ, j_i, S_{ij})	An unrestricted pullback of arbitrary tensor distributions to Cauchy hypersurfaces
Einstein constraint projections and gravitational ADM Legendre structure	A matter Hamiltonian inferred from stress-energy alone
Source-fibre-preserving local prequotient covariant–canonical reconstruction	A global reduced phase space or identification of all refoliations as gauge
Compact-core multiplier classification and conditional boundary-charge classification	A denial of physical boundary time or relational clocks
The disjoint exhaustive vacuum/canonical/forced handoff family G_1	Any Wheeler–DeWitt operator, ordering, regularization, anomaly, domain, or inner-product result

The calculation uses familiar ADM geometry. The new theorem object is not the historical priority of those identities. It is the dependency-preserving interface connecting the AASC kernel, realized physical interior, finite transport theorem, minimal Einstein projection, attachment-safe source representation, canonical reconstruction, boundary standing, and the precisely bounded handoff to later Wheeler–DeWitt papers.

Part I

Claim lock, dependency order, and frozen target

1. Exact theorem act and novelty boundary

1.1 The theorem act

Definition 1.1 (WDW-1 theorem act). The WDW-1 theorem act is the construction and report of a canonical projection of one already accepted Einstein-source theorem object on the fixed target $\mathcal{T}_{W1}^0$. Its proof object contains: (i) imported AASC necessity and projection theorems; (ii) conventional differential-geometric and Dirac–Bergmann calculations; and (iii) new interface certificates showing that the connection, source, canonical variables, boundary data, and reportability claims occupy the same target.

The no-backflow order is

$$\begin{aligned}
& \text{TargetAdeq}_D \implies K_D = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}) \implies A_D^+ \implies I_D, \\
& \text{CovDiffTarget}(D) \implies \text{TC}_D \implies \nabla \text{ is a connection,} \\
& \text{MinMetricNat}_1(M, g) \implies \nabla = \nabla^{\text{LC}}, \\
& I_D \implies \text{Gravity}_D \implies (M, g, \nabla^{\text{LC}}, T_{\mu\nu}^{\text{closed}}), \\
& \text{frozen covariant realization} \implies \text{ADM prequotient package,} \\
& \exists(\mathfrak{M}, \text{Src}_{\mathfrak{M}}) \mathfrak{C}_{\text{src}}^+(\mathfrak{M}, \text{Src}_{\mathfrak{M}}) \implies \text{total matter-coupled constraint package.}
\end{aligned}$$

No arrow reverses. ADM variables do not authorize the Einstein projection, the metric does not authorize the kernel, the stress tensor does not create its own upstream attachment, and a later quantum construction cannot repair a failed classical gate.

1.2 Novelty and historical boundary

This paper does not claim the first $3 + 1$ decomposition, the first ADM Hamiltonian, the first Dirac analysis of generally covariant systems, or the first bottom-up reconstruction of geometrodynamics [1–3]. It does not derive the Wheeler–DeWitt equation. Its contribution is the exact AASC-native ancestry theorem, including:

1. connection closure before foliation;
2. source attachment and Cauchy-trace discipline;
3. separation of covariant source projections from a canonical matter lift;
4. a prequotient reconstruction theorem that does not preclaim refoliation closure;
5. a boundary-sensitive classification of lapse, shift, and temporal data; and
6. an audit-complete handoff to the spatial-quotient and hypersurface-deformation gates.

2. Proof-class lock and correctness-relevant criticism

Definition 2.1 (Correctness-relevant criticism). A criticism is correctness-relevant to WDW-1 when it preserves the frozen target and identifies at least one of the following: a failed target instantiation; a failed imported theorem or hypothesis; a failed branch occupation; an incorrect $3 + 1$ identity; an invalid source projection; an invalid Legendre or reconstruction step; a nondifferentiable boundary generator; or a report stronger than the constructed theorem object.

Proposition 2.2 (Proof-class exhaustiveness). *Every correctness-relevant criticism of the WDW-1 endpoint attacks an import, a conventional calculation, an interface certificate, or the reportability boundary. A request that changes dimension, signature, connection class, source regularity, topology, boundary class, matter content, or quantum target is a target update rather than a counterexample to $\mathcal{T}_{\text{W1}}^0$.*

Proof. The theorem act is the conjunction of the four named proof carriers. If the target is preserved, a false conclusion must arise from a false imported premise, an invalid conventional inference, a failed interface instantiation, or an overstatement of the conjunction. If the proposed alternative changes a coordinate of $\mathcal{T}_{\text{W1}}^0$, it no longer instantiates the same antecedent. This is a scope classification, not an immunity claim: every preserved-target attack remains admissible and must be answered at its named link. $\square$

3. Frozen target and target-update discipline

Definition 3.1 (Core target). The WDW-1 core target is

$$\mathcal{T}_{W1}^0 = (D, I_D, M, g, \nabla, \Sigma, \mathcal{F}, \hat{x}, T, J, \kappa, \Lambda, \mathcal{B}_0, \mathcal{R}, \mathcal{P}),$$

with:

1. one fixed kernel-faithful domain D and its singleton realized physical interior I_D ;
2. a smooth connected paracompact Hausdorff four-manifold M ;
3. a smooth time-oriented Lorentzian metric g of signature $(-+++)$;
4. a connection occupant ∇ to be fixed by the covariant-differentiation branch;
5. a smooth compact Cauchy three-manifold Σ without boundary and an admissible global splitting $\mathcal{F} : M \simeq \mathbb{R} \times \Sigma$;
6. an attachment-certified structured source class $\hat{x} \in \hat{Q}^{\text{src}}$, with unique symmetric smooth representative $T = T_{\hat{x}}$;
7. the closed exchange coordinate $J = 0$;
8. declared constants $\kappa \neq 0$ and Λ , neither numerically generated by the kernel;
9. compact boundaryless class $\mathcal{B}_0$;
10. regularity data $\mathcal{R}$ sufficient for all derivatives, restrictions, variations, and local reconstruction used below; and
11. the prequotient presentation comparison class $\mathcal{P}$.

For definiteness, the main proof may be read in the C^∞ category. A finite-regularity variant is valid only after replacing every appeal to smooth splitting, curvature, and Hamiltonian variation by a theorem at the stated Sobolev or Hölder order. This manuscript does not silently make that replacement.

Definition 3.2 (Extensions). The boundary and source-canonical extensions are

$$\mathcal{T}_{W1}^\partial = \mathcal{T}_{W1}^0 + (\partial M, \mathcal{B}_\partial, H_\partial, Q_\partial), \quad \mathcal{T}_{W1}^{\text{can-} \text{src}} = \mathcal{T}_{W1}^0 + (\mathfrak{M}, \text{Src}\mathfrak{M}, \mathfrak{C}_{\text{src}}^+(\mathfrak{M}, \text{Src}\mathfrak{M})).$$

A distributional extension additionally replaces the smooth source clause by a proved pullback or trace certificate. Each addition is a target update, not a reinterpretation of the core.

Definition 3.3 (Connection-occupied core subtype). The connection-occupied core is the certified subtype

$$\mathcal{T}_{W1}^{0, \nabla} = \left\{ \mathcal{T} \in \mathcal{T}_{W1}^0 : \mathfrak{C}_{\text{transport}}(\mathcal{T}) \wedge \mathfrak{C}_{\nabla}(\mathcal{T}) \wedge \text{MinMetricNat}_1(M, g) \right\}.$$

Membership records rather than infers the smooth standing-preserving finite-transport realization, every coordinate of the connection-occupancy certificate, and occupation of the minimal first-order natural metric comparison class. Theorems W1.1 and W1.8 apply only to this populated subtype.

Audit condition 3.4 (No silent target motion). Every theorem below is tagged as core, conditional source-canonical, or conditional boundary. Torsion, nonmetricity, independent connection, tetrads, spin connections, internal gauge constraints, fermions, singular strata, noncompact leaves, topology change, and quantum operators are excluded from the core unless a displayed target-update statement admits them.

4. Imported theorem ledger and no-backflow order

The corpus imports are used as theorem objects rather than as slogans. Table 1 records the exact source theorem or source label, the local conclusion, and the promotion that is prohibited. Full hypotheses and source lineage are repeated in Appendix B.

Table 1: Imported theorem-use ledger.

Source theorem object	WDW-1 use	Prohibited promotion
Kernel, “Main fixed-domain exhaustion theorem” (<code>thm:main-fixed-do-main-aasc</code>), “Scope-preserving invariance” (<code>thm:scope-preserving-invariance</code>), “Transport closure” (<code>thm:transport-closure</code>), and quotient identity [4]	Target adequacy forces K_D ; same-domain continuation and lawful redescription preserve the invariant bundle.	No metric, manifold, Cauchy split, source, or Hamiltonian follows from the kernel alone.
Unus, “No repairs” (<code>thm:no-repairs</code>), “No generators” (<code>thm:no-generators</code>), “No illicit scope transport” (<code>lem:no-scope-transport</code>), and “Uniqueness of the admissible interior” (<code>thm:unique</code>) [5]	Blocks reverse authorization, deferred repair, and plural same-domain interiors.	Does not imply a unique foliation, boundary condition, or matter realization.
Realized Physical Interior, local AER forcing and gravity-role theorems [6]	Supplies the finite/local coisotropic, characteristic, first-class, and multiplier role architecture and the gravity role upstream of metric projection.	Is not a global analytic theorem for the ADM field phase space.
Einstein projection, “AASC gravity necessity,” “No primitive metric backflow,” “No hidden nonmetric transport,” and the minimal first-response Einstein closure [7]	Freezes the four-dimensional Lorentzian Einstein-source theorem object.	Does not construct the ADM phase space, source canonical lift, or quantum gravity.

Source theorem object	WDW-1 use	Prohibited promotion
Module A, Theorems 5.1, 6.4–6.5, 7.2–7.5, Example 13.2, Proposition 13.3, and Ward/composite closure [8]	Supplies attachment-safe source representation, uniqueness, naturality, smooth subtype, the attached zero and Klein–Gordon occupants, and the closed Ward ledger.	A tensor distribution has no automatic Cauchy trace or matter Hamiltonian.
Covariant Differentiation, Theorems 11.1, 13.1, 17.2, and 19.1 [9]	Supplies identity principal symbol, connection normal form, and Levi–Civita closure on MinMetricNat_1 .	No connection on superspace, functional derivative, ordering, or operator domain follows.
Non-Quantizability, Theorems 2.10 and 5.3 [10]	Blocks a primitive same-domain $\text{GR} \rightarrow \text{QM}$ map and fixes the later realization-certificate burden.	Does not deny effective quantization or construct a quantum realization.
Physical Clock, Theorem <code>thm:no-external-time</code> , relational descent, and lapse/shift boundary guard [11]	Blocks undeclared same-scope external-clock import; retains relational and boundary time when certified.	Does not prove the ADM deformation algebra, a global clock, or a Hilbert space.
Arrow and continuation-order papers [12, 13]	Irreversibility is non-temporal upstream; temporal and causal use is downstream continuation structure.	Does not make coordinate t physical or derive Lorentzian causal geometry.
Unification capstone, Theorems 10.1, 11.1, 14.1, 21.1, and 82.1 [14]	Freezes AER, Einstein, Module-A, paired-projection, and reportability imports.	Explicitly does not report all interacting quantum-gravity dynamics.

Theorem 4.1 (Import consistency). *The imported conclusions used in WDW-1 have compatible local instantiations on $\mathcal{T}_{\text{W1}}^0$, and no import is used beyond its declared conclusion.*

Proof. The kernel and Unus objects govern fixed-domain standing and dependency. The RPI/AER import is used only for the regular finite/local role architecture; no global ADM quotient is inferred. The Einstein import fixes a smooth four-dimensional Lorentzian metric-source equation, while Module A fixes the source attachment and smooth subtype. The covariant-differentiation theorem is applied to the finite-rank tangent and tensor bundles of that spacetime, not to an infinite-dimensional configuration space. Clock, arrow, and non-quantizability theorems constrain reportability but supply no missing ADM identity. The new assumptions of global hyperbolicity, compact Cauchy topology, and smooth source trace are explicit coordinates of $\mathcal{T}_{\text{W1}}^0$. Thus the conclusions compose in the displayed forward order and none is used to authorize an upstream premise. $\square$

5. Why WDW-1 is not primitive quantization

The type distinction imported from the non-quantizability theorem is

$$\text{GR} \not\rightarrow \text{QM} \quad \text{as a primitive same-domain foundational map.}$$

WDW-1 does something narrower. It constructs a classical canonical presentation of the accepted Einstein projection so that a later constraint-native joint realization has a well-typed classical interface.

The familiar formal substitution

$$\pi^{ij} \mapsto -i\hbar \frac{\delta}{\delta h_{ij}}$$

does not occur as a theorem step here. Representation choice, observable assignment, constraint closure, physical-state selection, measure or inner product, boundary extraction, and continuation are later standing-bearing divergence loci whenever they change the physical solution space. Calling them “conventions” does not discharge them.

Part II

Connection ancestry and spacetime split

6. Occupation of the covariant-differentiation target

Definition 6.1 (Connection-occupancy certificate). The certificate

$$\mathfrak{C}_{\nabla} = (TM, g, \text{Local}, \text{DiffNat}, j^1g, \nabla g = 0, \text{NoExtraSel})$$

records: (i) TM and its natural tensor bundles as finite-rank carriers; (ii) g as the only connection-selecting metric-side tensor; (iii) locality; (iv) diffeomorphism naturality; (v) dependence on at most the first jet of g ; (vi) metric-comparison preservation; and (vii) absence of orientation, foliation, source, boundary, curvature, nonlocal, or auxiliary selectors.

The finite-transport theorem first yields the connection *type*. Smooth standing-preserving transport on the standard finite linear branch has an identity principal symbol and infinitesimalizes to an operator satisfying

$$\nabla_X(fs) = X(f)s + f\nabla_Xs, \quad \nabla_{fX+gY}s = f\nabla_Xs + g\nabla_Ys.$$

In a local frame, $\nabla = d + A$. This does not yet determine the coefficients A . Metric comparison preservation then gives $\nabla g = 0$. Numerical Levi–Civita closure requires the narrower natural first-order branch described next.

7. Levi–Civita closure before foliation

Definition 7.1 (Minimal first-order natural metric branch). $\text{MinMetricNat}_1(M, g)$ is the comparison class of connections on TM selected locally and diffeomorphism-naturally from j^1g , preserving g , and using no additional standing-bearing selector.

Proposition 7.2 (Certified minimal-metric occupancy). *For every $\mathcal{T} \in \mathcal{T}_{W1}^{0,\nabla}$, the predicate*

$$\text{MinMetricNat}_1(M, g)$$

is populated, and each of its seven occupancy coordinates has a named upstream or frozen-target witness.

Proof. This is the subtype predicate of Definition 3.3, unpacked against the source clauses in Table 2. The accepted Einstein theorem object supplies the tangent and natural tensor carriers on (M, g) . The smooth finite-transport realization and Covariant Differentiation Theorems 11.1, 13.1, 17.2, and 19.1 supply connection type, identity principal symbol, and the minimal-branch conclusion. The frozen minimal first-response clauses supply locality, naturality, first-jet dependence, metric comparison, and the no-extra-selector condition. Nothing in this argument derives subtype membership from the bare tuple of Definition 3.1; membership is a checked hypothesis. $\square$

Table 2: Connection-occupancy import map.

Occupancy coordinate	Witness
Smooth standing-preserving finite transport	Frozen $\mathfrak{C}_{\text{transport}}$ realization and Covariant Differentiation Theorem 19.1 hypotheses
Identity principal symbol	Covariant Differentiation Theorem 11.1
Connection normal form	Covariant Differentiation Theorem 13.1 and Theorem 19.1
Metric comparison preservation	Covariant Differentiation Lemma 16.2, instantiated by the frozen metric comparison
Locality and first-jet dependence	MinMetricNat ₁ clauses 3 and 5 in Covariant Differentiation Definition 17.1
Diffeomorphism naturality	MinMetricNat ₁ clause 4
No additional selector	MinMetricNat ₁ clauses 2 and 7 and the fixed WDW-1 target ledger

Lemma 7.3 (Difference-tensor vanishing). *Let both ∇ and ∇^{LC} occupy MinMetricNat₁(M, g). Their difference*

$$C(X, Y) = \nabla_X Y - \nabla_X^{\text{LC}} Y$$

is a (1, 2)-tensor natural in $j^1 g$. On the declared branch $C = 0$.

Proof. The tensoriality of a difference of connections is standard. The imported Natural Difference-Tensor Vanishing Theorem applies because C is local, first-jet, diffeomorphism-natural, metric-compatible, and has no additional selector. In normal coordinates for g at a point the first jet vanishes; naturality leaves no nonzero tensorial value constructible at that order. Hence C vanishes pointwise. $\square$

Theorem W1. 1 (Dependency and branch-occupancy theorem). For every $\mathcal{T} \in \mathcal{T}_{\text{W1}}^{0, \nabla}$, the accepted Einstein-source realization is downstream of I_D , and the connection occupant is fixed before foliation by

$$\nabla = \nabla^{\text{LC}}.$$

Torsion, nonmetricity, independent-connection, tetrad, teleparallel, higher-jet, nonlocal, or boundary-selected alternatives are changed targets rather than same-branch competitors.

Proof. Import consistency gives the forward kernel–interior–Einstein dependency. Membership in $\mathcal{T}_{\text{W1}}^{0, \nabla}$ and Proposition 7.2 establish the populated connection-occupancy certificate. Lemma 7.3, equivalently the imported Theorems 17.2 and 19.1 of *Covariant Differentiation from the AASC Kernel*, yields the displayed equality. Since no foliation datum appears in either certificate or proof, the result precedes and cannot be selected by the 3 + 1 presentation. $\square$

8. Global hyperbolicity and admissible foliation

Global hyperbolicity is a frozen projection anchor, not a consequence inferred here from the four kernel roles. By the smooth splitting theorem [15, 16], the target admits a smooth Cauchy temporal function t and a diffeomorphism

$$M \simeq \mathbb{R} \times \Sigma, \quad \Sigma_t = \{t\} \times \Sigma,$$

with every Σ_t a spacelike Cauchy hypersurface. The metric takes the ADM form

$$g = -N^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt), \quad N > 0. \quad (1)$$

Writing $t^\mu = (\partial/\partial t)^\mu$,

$$t^\mu = N n^\mu + N^\mu, \quad n_\mu dx^\mu = -N dt, \quad n^\mu = \frac{1}{N}(t^\mu - N^\mu),$$

where $N^\mu n_\mu = 0$. Time orientation selects the future unit normal n^μ ; it does not select a unique foliation.

Proposition 8.1 (Foliation non-selection). *Existence of a smooth global Cauchy foliation on $\mathcal{T}_{W1}^0$ does not confer physical privilege on one such foliation.*

Proof. The splitting theorem is existential. If t is Cauchy temporal, many strictly monotone reparameterizations and, generally, many inequivalent Cauchy temporal functions remain available. Neither the target tuple nor Theorem W1.1 supplies a selector among them. Selecting one foliation is therefore presentation data used to perform the split. The stronger claim that all appropriate refoliations lie in the physical characteristic kernel requires the hypersurface-deformation and quotient analysis allocated to WDW-3. $\square$

9. ADM kinematics and convention lock

The convention ledger is:

$$n^\mu n_\mu = -1, \quad h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu, \quad h^\mu{}_\nu = \delta^\mu{}_\nu + n^\mu n_\nu.$$

The intrinsic Levi–Civita derivative on each leaf is denoted D_i . The positive-sign extrinsic curvature and normal acceleration are

$$K_{ij} = h_i{}^\mu h_j{}^\nu \nabla_\mu n_\nu = \frac{1}{2N} (\dot{h}_{ij} - \mathcal{L}_{\vec{N}} h_{ij}), \quad a_i = D_i \ln N. \quad (2)$$

Thus

$$\dot{h}_{ij} = 2N K_{ij} + D_i N_j + D_j N_i, \quad \sqrt{-g} = N \sqrt{h}.$$

In coordinates adapted to the split,

$$g_{\mu\nu} = \begin{pmatrix} -N^2 + N_k N^k & N_j \\ N_i & h_{ij} \end{pmatrix}, \quad g^{\mu\nu} = \begin{pmatrix} -N^{-2} & N^j N^{-2} \\ N^i N^{-2} & h^{ij} - N^i N^j N^{-2} \end{pmatrix}.$$

The Riemann convention is

$$[\nabla_\mu, \nabla_\nu] v^\rho = R^\rho{}_{\sigma\mu\nu} v^\sigma, \quad R_{\sigma\nu} = R^\rho{}_{\sigma\rho\nu}.$$

All later signs follow from these displayed choices. A source or reference using $K_{ij} = -\frac{1}{2} \mathcal{L}_n h_{ij}$ must be translated before comparison.

10. Gauss–Codazzi–Ricci decomposition

For spatial vectors, the Gauss equation in the frozen convention is

$$h_a{}^\alpha h_b{}^\beta h_c{}^\gamma h_d{}^\delta R_{\alpha\beta\gamma\delta} = {}^{(3)}R_{abcd} + K_{ac} K_{bd} - K_{ad} K_{bc}. \quad (3)$$

The Codazzi projection and twice-contracted Gauss equation give

$$h_i{}^\nu n^\mu G_{\mu\nu} = D_j K^j{}_i - D_i K, \quad (4)$$

$$2G_{\mu\nu} n^\mu n^\nu = {}^{(3)}R + K^2 - K_{ij} K^{ij}. \quad (5)$$

The scalar curvature decomposition is

$${}^{(4)}R = {}^{(3)}R + K_{ij} K^{ij} - K^2 + 2\nabla_\mu (n^\mu K - a^\mu). \quad (6)$$

The last term is a divergence; whether it vanishes, is cancelled by an action surface term, or becomes a charge is a boundary question and is never discarded without a boundary ledger.

For later reconstruction it is useful to state the spatial evolution equation explicitly. The spatial projection of the Einstein equation is equivalent, together with the constraints, to

$$(\partial_t - \mathcal{L}_{\vec{N}})K_{ij} = D_i D_j N - N \left[{}^{(3)}R_{ij} + K K_{ij} - 2K_i^k K_{kj} - \Lambda h_{ij} - \kappa \left(S_{ij} - \frac{1}{2} h_{ij} (S - \rho) \right) \right], \quad (7)$$

where $S = h^{ij} S_{ij}$. This sign is consistent with (2); changing the extrinsic-curvature convention changes the entire line, not one isolated term.

Lemma 10.1 (Projection completeness). *Let $E_{\mu\nu}$ be a symmetric tensor. Relative to n^μ , it is uniquely determined by*

$$E_{nn} = E_{\mu\nu} n^\mu n^\nu, \quad E_{ni} = E_{\mu\nu} n^\mu h_i^\nu, \quad E_{ij} = E_{\mu\nu} h_i^\mu h_j^\nu.$$

Hence $E_{nn} = E_{ni} = E_{ij} = 0$ if and only if $E_{\mu\nu} = 0$.

Proof. Substitute $g^\mu{}_\nu = h^\mu{}_\nu - n^\mu n_\nu$ in both slots of E . The resulting normal, mixed, and spatial pieces reconstruct E uniquely. The ten independent components split as $1 + 3 + 6$. $\square$

Part III

Attachment-safe source descent

11. Module-A source ancestry

Module A begins with an upstream response quotient Q_{resp} . Only the subtype $Q_{\text{resp}}^{\text{src}}$ carrying a complete metric-incidence attachment certificate is source-bearing. Its structured quotient retains the response presentation, attachment certificate, and complete operational witness:

$$\widehat{Q}^{\text{src}} = \{(p, a_p, w_p) : \text{SrcBear}_g(p, a_p, w_p)\} / \sim_{\text{src}}.$$

For $x \in \widehat{Q}^{\text{src}}$, the witness W_x is represented uniquely by a symmetric tensor-density distribution,

$$W_x(\delta g) = \frac{1}{2} \langle T_x, \delta g \rangle.$$

The representation is natural and faithful under the declared lawful-redescription and boundary equivalence. The graph-equality clause is essential: the witness reproduces the independently fixed upstream metric-response relation; it does not create source standing by being written beside a response.

Proposition 11.1 (No canonical promotion by source label). *Writing $T_{\mu\nu}$, ρ , or $\mathcal{H}_\perp^m$ beside an upstream response constructs neither a source attachment nor a canonical matter lift.*

Proof. Module-A attachment requires graph equality with the upstream response and complete probe faithfulness. A canonical lift additionally requires variables, momenta, a symplectic form, a Legendre or equivalent construction, constraint generators, and matching theorems. The symbols alone supply none of these witnesses. $\square$

12. Cauchy-admissible source certificate

Certificate 12.1 (Smooth core source). For a source T , define

$$\mathfrak{C}_\Sigma(T) = (\text{Reg}_T, \text{Trace}_\Sigma, \rho, j, S, \text{WardSplit}, \text{BoundaryStatus}).$$

The core occupant satisfies:

1. $x \in \widehat{Q}^{\text{src}}$ has the Module-A smooth density certificate;
2. its representative is a smooth symmetric covariant tensor $T_{\mu\nu}$;
3. restriction and contraction with the smooth fields n^μ and $h^\mu{}_\nu$ are defined on every leaf;
4. the projected fields vary smoothly with t ;
5. the covariant closed Ward identity $\nabla_\mu T^{\mu\nu} = 0$ holds; and
6. $\partial\Sigma = \emptyset$, so no spatial-boundary flux term is omitted.

The smooth core is deliberately narrower than Module A’s full distributional image. For a distribution T , a pullback i_t^*T along $i_t : \Sigma_t \hookrightarrow M$ exists only under an explicit trace theorem. A standard sufficient microlocal condition is

$$\text{WF}(T) \cap N^*\Sigma_t = \emptyset,$$

with the tensor contractions and t -dependence controlled in the chosen distribution topology [17]. WDW-1 records that extension in Appendix H; it does not use it in the core theorem.

13. Source 3 + 1 decomposition

Define the source projections

$$\rho = T_{\mu\nu}n^\mu n^\nu, \quad j_i = -h_i^\mu n^\nu T_{\mu\nu}, \quad S_{ij} = h_i^\mu h_j^\nu T_{\mu\nu}. \quad (8)$$

With $j_\mu n^\mu = S_{\mu\nu}n^\mu = 0$, the tensor reconstructs as

$$T_{\mu\nu} = \rho n_\mu n_\nu + n_\mu j_\nu + n_\nu j_\mu + S_{\mu\nu}. \quad (9)$$

Proposition 13.1 (Source split uniqueness and descent). *On the smooth Cauchy-admissible subtype, (ρ, j_i, S_{ij}) is uniquely determined by the same Module-A representative, transforms naturally under leaf-preserving spatial redescription, inherits support and boundary status, and reconstructs $T_{\mu\nu}$ by (9).*

Proof. The normal/tangential projectors form a pointwise direct splitting of the symmetric tensor bundle. Contractions (8) therefore give the unique coefficients. Module-A naturality transports T , while n and h transport with the split, so the coefficients transform as one scalar, one spatial covector, and one symmetric spatial tensor. Restriction and contraction do not enlarge support. A boundary-supported component cannot be removed by this split and remains in the boundary coordinate. $\square$

14. Ward/divergence split

Permit temporarily a declared exchange current

$$\nabla_\mu T^{\mu\nu} = J^\nu, \quad \varepsilon = -n_\nu J^\nu, \quad q_i = h_{i\nu} J^\nu.$$

Using $\nabla_\mu n_\nu = K_{\mu\nu} - n_\mu a_\nu$, the normal projection is

$$\mathcal{L}_n \rho + D_i j^i + K \rho + K_{ij} S^{ij} + 2a_i j^i = \varepsilon. \quad (10)$$

The spatial projection is

$$(\mathcal{L}_n j)_i + K j_i + D_j S^j{}_i + \rho a_i + S_i{}^j a_j = q_i. \quad (11)$$

Here $(\mathcal{L}_n j)_i$ denotes the spatial pullback of the spacetime Lie derivative. If instead a projected covariant normal derivative is used, the term $K_i{}^j j_j$ must be displayed separately. Equations (10)–(11) are therefore not notation-free identities; their derivative convention is part of the sign ledger.

Lemma 14.1 (Ward recombination). *On the smooth subtype, equations (10) and (11) recombine uniquely to $\nabla_\mu T^{\mu\nu} = J^\nu$.*

Proof. The two displayed equations are respectively $-n_\nu$ and $h_{i\nu}$ applied to the covariant divergence. Since $\delta^\nu_\alpha = h^\nu_\alpha - n^\nu n_\alpha$, normal and spatial projections span the tangent space. Their conjunction is therefore equivalent to the original vector equation. Direct substitution of (9) gives the extrinsic-curvature and acceleration terms shown. $\square$

Theorem W1. 2 (Cauchy-source projection theorem). Every source in the declared smooth attachment-certified Cauchy-admissible subtype has unique projections (ρ, j_i, S_{ij}) descending from the same Module-A tensor representative. Their energy and momentum balances are exactly the normal and spatial projections of the same Ward/exchange ledger. No distributional or canonical matter structure is inferred.

Proof. Certificate 12.1 licenses restriction and contraction. Proposition 13.1 proves uniqueness, descent, and reconstruction. Lemma 14.1 proves exact Ward equivalence. The final sentence follows because neither the smooth trace certificate nor a tensor-bundle split supplies a canonical model, source-realization map, or the coordinates of $\mathfrak{C}_{\text{src}}^+$. $\square$

Part IV

Constraint projection and canonical construction

15. Einstein constraint projections

The accepted metric equation is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}. \quad (12)$$

Contracting twice with n^μ , using $g_{nn} = -1$ and (5), gives

$${}^{(3)}R + K^2 - K_{ij}K^{ij} - 2\Lambda = 2\kappa\rho. \quad (13)$$

Contracting with $n^\mu h_i{}^\nu$, using $g_{ni} = 0$, $T_{ni} = -j_i$, and (4), gives

$$D_j(K^j{}_i - \delta_i^j K) = -\kappa j_i. \quad (14)$$

These statements use no matter phase-space variable. They are the normal and tangential geometric projections of the same covariant equation.

Theorem W1. 3 (Einstein constraint-projection theorem). On every leaf of $\mathcal{T}_{\text{W1}}^0$, the normal-normal and normal-tangential projections of (12) are exactly (13) and (14), with source data descending from the Module-A representative.

Proof. Equation (13) follows by substituting (5) in $G_{nn} - \Lambda = \kappa\rho$. Equation (14) follows by substituting (4) in $G_{ni} = -\kappa j_i$. The source ancestry is Theorem W1.2. No canonical matter term appears in the proof. $\square$

16. Downstream Einstein–Hilbert representation

The action is used after the Einstein theorem object has been fixed. It is a downstream equivalent representation on the action-presentable smooth branch, not the primitive source of the AASC gravity role:

$$S_g[g] = \frac{1}{2\kappa} \int_M \sqrt{-g} ({}^{(4)}R - 2\Lambda) d^4x + S_\partial. \quad (15)$$

For a compact slab bounded by two spacelike Cauchy leaves, S_∂ includes the Gibbons–Hawking–York term with the orientation dictated by the endpoint normals [18, 19]. There is no timelike spatial

boundary in the core. Substitution of (6), together with the boundary completion, yields the bulk ADM Lagrangian

$$L_g = \int_{\Sigma} \mathcal{L}_g d^3x, \quad \mathcal{L}_g = \frac{N\sqrt{h}}{2\kappa} \left({}^{(3)}R + K_{ij}K^{ij} - K^2 - 2\Lambda \right). \quad (16)$$

Proposition 16.1 (Action compatibility). *On the smooth action-presentable core with fixed induced endpoint metric, the variation of (15) gives the same vacuum-side tensor $G_{\mu\nu} + \Lambda g_{\mu\nu}$ used in (12); coupling to an action-presentable source reproduces the same Module-A tensor class.*

Proof. The gravitational variation with the GHY endpoint term removes the normal derivative of δg and leaves

$$\delta S_g = \frac{1}{2\kappa} \int_M \sqrt{-g} (G_{\mu\nu} + \Lambda g_{\mu\nu}) \delta g^{\mu\nu} d^4x.$$

Module-A Hilbert compatibility identifies the source variation with the already attached tensor witness on the action-presentable subclass. Thus the action reproduces, rather than authorizes, the frozen covariant equation. $\square$

17. ADM Legendre map

Differentiating (16) with respect to $\dot{h}_{ij}$ gives

$$\pi^{ij} = \frac{\partial \mathcal{L}_g}{\partial \dot{h}_{ij}} = \frac{\sqrt{h}}{2\kappa} (K^{ij} - K h^{ij}). \quad (17)$$

The lapse and shift carry no velocities:

$$P_N = \frac{\partial \mathcal{L}_g}{\partial \dot{N}} = 0, \quad P_i = \frac{\partial \mathcal{L}_g}{\partial \dot{N}^i} = 0.$$

Taking the trace of (17) and inverting on nondegenerate positive-definite h gives

$$\pi = -\frac{\sqrt{h}}{\kappa} K, \quad K_{ij} = \frac{2\kappa}{\sqrt{h}} \left(\pi_{ij} - \frac{1}{2} \pi h_{ij} \right). \quad (18)$$

The coordinate shorthand for the weak gravitational two-form defined below is

$$\Omega_g = \int_{\Sigma} \delta \pi^{ij} \wedge \delta h_{ij} d^3x. \quad (19)$$

Equation (19) is interpreted by the bilinear formula (23), not as a claim about an undeclared infinite-dimensional quotient.

The Legendre transform and one integration by parts on compact Σ yield

$$H_g[N, N^i] = \int_{\Sigma} \left(N \mathcal{H}_{\perp}^g + N^i \mathcal{H}_i^g \right) d^3x, \quad (20)$$

$$\mathcal{H}_{\perp}^g = \frac{2\kappa}{\sqrt{h}} \left(\pi_{ij} \pi^{ij} - \frac{1}{2} \pi^2 \right) - \frac{\sqrt{h}}{2\kappa} ({}^{(3)}R - 2\Lambda), \quad (21)$$

$$\mathcal{H}_i^g = -2D_j \pi^j_i. \quad (22)$$

18. Smooth gravitational phase space, domains, and constraint map

Definition 18.1 (Smooth gravitational canonical carrier). For the compact smooth Cauchy leaf Σ , define

$$\mathcal{P}_g^{\infty} = \text{Met}^{\infty}(\Sigma) \times \Gamma^{\infty} \left(S^2 T\Sigma \otimes \text{Dens}(\Sigma) \right).$$

Its points are (h_{ij}, π^{ij}) , where h is a smooth Riemannian metric and π^{ij} a smooth symmetric contravariant density of weight one. On tangent pairs $(\delta h_A, \delta \pi_A)$, set

$$\Omega_g((\delta h_1, \delta \pi_1), (\delta h_2, \delta \pi_2)) = \int_{\Sigma} \left(\delta \pi_1^{ij} \delta h_{2ij} - \delta \pi_2^{ij} \delta h_{1ij} \right). \quad (23)$$

This is the weak canonical two-form on the smooth prequotient carrier. The Hamiltonian convention is

$$\iota_{X_H} \Omega_g = -\delta H, \quad \dot{h}_{ij} = \frac{\delta H}{\delta \pi^{ij}}, \quad \dot{\pi}^{ij} = -\frac{\delta H}{\delta h_{ij}}.$$

Definition 18.2 (Smearing and functional domains). The positive lapse space used to reconstruct a time-oriented Lorentzian metric is

$$\mathcal{N}_+ = C_+^\infty(\Sigma).$$

The linear smearing spaces used to test constraint candidates are

$$\mathcal{N} = C^\infty(\Sigma), \quad \mathcal{X} = \Gamma^\infty(T\Sigma).$$

Let $\mathcal{A}_{\text{reg}}(\mathcal{P}_g^\infty)$ denote the smooth regular functionals whose first variational derivatives are smooth fields of the dual density type and whose integrations by parts are licensed by compactness of Σ . Every Poisson bracket or Hamiltonian vector-field statement in WDW-1 is restricted to this declared domain, or to a separately named common matter domain.

Definition 18.3 (Gravitational constraint map and vacuum locus). Define

$$\Phi_g : \mathcal{P}_g^\infty \longrightarrow \Gamma^\infty(\text{Dens } \Sigma) \oplus \Gamma^\infty(T^*\Sigma \otimes \text{Dens } \Sigma)$$

by

$$\Phi_g(h, \pi) = (\mathcal{H}_\perp^g(h, \pi), \mathcal{H}_i^g(h, \pi)), \quad \mathcal{C}_{\text{vac}} = \Phi_g^{-1}(0).$$

The zero locus is prequotient: WDW-1 does not yet identify the characteristic distribution, prove a global quotient, or complete the smooth carrier in a Sobolev topology.

19. Extended Dirac phase space and multiplier reduction

The extended coordinates are

$$(h_{ij}, \pi^{ij}; N, P_N; N^i, P_i).$$

The primary Hamiltonian is

$$H_P = H_g[N, N^i] + \int_{\Sigma} (\lambda_N P_N + \lambda^i P_i) d^3x. \quad (24)$$

Preservation of the primary constraints gives

$$\dot{P}_N = -\mathcal{H}_\perp^g \approx 0, \quad \dot{P}_i = -\mathcal{H}_i^g \approx 0$$

in vacuum. With a certified canonical source lift, the secondary expressions are replaced by the total constraints of Section 26. With an attachment-safe covariant source but no canonical lift, the geometric source equations (13)–(14) constrain the gravitational data, but no total matter Hamiltonian is asserted.

Proposition 19.1 (Extended/reduced agreement). *Eliminating the primary multiplier sector $(N, P_N; N^i, P_i)$ from the extended Dirac presentation leaves the reduced multiplier presentation $(h_{ij}, \pi^{ij}; N, N^i)$ with N, N^i varied as multipliers.*

Proof. The primary momenta vanish and their arbitrary multipliers change no equation for (h, π) . Variation of the reduced action

$$\int dt \int_{\Sigma} (\pi^{ij} \dot{h}_{ij} - N \mathcal{H}_\perp^g - N^i \mathcal{H}_i^g) d^3x$$

with respect to N, N^i reproduces the same secondary equations as preservation of P_N, P_i . The reduction removes the primary bookkeeping pairs; it does not yet quotient the secondary characteristic flows. $\square$

Theorem W1. 4 (Gravitational Legendre and multiplier theorem). On the compact boundaryless core, the downstream Einstein–Hilbert representation has the Legendre map (17)–(18), gravitational Hamiltonian (20), and primary lapse/shift constraints in the extended presentation. Lapse and shift are multipliers in the reduced presentation. The two presentations agree after eliminating the primary multiplier sector.

Proof. Equations (17)–(22) are the explicit Legendre calculation. Equation (24) gives the primary constraints and their preservation. Proposition 19.1 proves the comparison. Compactness and $\partial\Sigma = \emptyset$ justify the integration by parts; boundary extensions require a separate differentiability certificate. $\square$

20. Constraint identity

Substituting (17) into the quadratic form gives

$$\pi_{ij}\pi^{ij} - \frac{1}{2}\pi^2 = \frac{h}{4\kappa^2}(K_{ij}K^{ij} - K^2).$$

Consequently

$$\mathcal{H}_\perp^g + \sqrt{h}\rho = 0 \iff {}^{(3)}R + K^2 - K_{ij}K^{ij} - 2\Lambda = 2\kappa\rho, \quad (25)$$

$$\mathcal{H}_i^g - \sqrt{h}j_i = 0 \iff D_j(K^j_i - \delta_i^j K) = -\kappa j_i. \quad (26)$$

The signs in the second line follow from $\mathcal{H}_i^m = -\sqrt{h}j_i$ on a certified matter lift, so the total equation is $\mathcal{H}_i^g + \mathcal{H}_i^m = 0$.

Theorem 20.1 (Constraint-identity theorem). *Under the frozen Legendre map, the canonical expressions*

$$\mathcal{H}_\perp^g + \sqrt{h}\rho, \quad \mathcal{H}_i^g - \sqrt{h}j_i$$

vanish exactly when the geometric Einstein constraints (13)–(14) hold.

Proof. The first equivalence is the displayed DeWitt quadratic identity multiplied by $-2\kappa/\sqrt{h}$. For the second,

$$-2D_j\pi^j_i = -\frac{\sqrt{h}}{\kappa}D_j(K^j_i - \delta_i^j K),$$

because $D_j\sqrt{h} = 0$. Adding $-\sqrt{h}j_i$ and setting the result to zero gives (14). $\square$

21. Canonical evolution and source forcing

The gravitational Hamilton equation for the metric is

$$\dot{h}_{ij} = \frac{4\kappa N}{\sqrt{h}} \left(\pi_{ij} - \frac{1}{2}\pi h_{ij} \right) + \mathcal{L}_{\vec{N}} h_{ij}, \quad (27)$$

which is exactly (2) after (18). The vacuum gravitational equation for the momentum is

$$\begin{aligned} \dot{\pi}^{ij} = & -\frac{N\sqrt{h}}{2\kappa} \left({}^{(3)}R^{ij} - \frac{1}{2}h^{ij}{}^{(3)}R + \Lambda h^{ij} \right) \\ & - \frac{4\kappa N}{\sqrt{h}} \left(\pi^i_k \pi^{kj} - \frac{1}{2}\pi \pi^{ij} \right) + \frac{\kappa N}{\sqrt{h}} h^{ij} \left(\pi_{kl} \pi^{kl} - \frac{1}{2}\pi^2 \right) \\ & + \frac{\sqrt{h}}{2\kappa} (D^i D^j N - h^{ij} D_k D^k N) + \mathcal{L}_{\vec{N}} \pi^{ij}. \end{aligned} \quad (28)$$

For a density of weight one,

$$\mathcal{L}_{\vec{N}} \pi^{ij} = D_k (N^k \pi^{ij}) - \pi^{ik} D_k N^j - \pi^{kj} D_k N^i.$$

An attachment-safe source without a populated source-family certificate enters the spatial Einstein equation as a covariantly prescribed forcing term. The term is calculated rather than defined by the desired answer. Differentiating (17), inserting the source part of (7), and taking the trace gives the intermediate contribution

$$\dot{\pi}^{ij} \Big|_{\text{src,raw}} = \frac{N\sqrt{h}}{2} (S^{ij} - \rho h^{ij}).$$

When the remaining geometrical expression is converted to the vacuum canonical right-hand side (28), use of the sourced Hamiltonian constraint contributes $+N\sqrt{h}\rho h^{ij}/2$. Hence

$$\boxed{\mathfrak{F}_T^{ij} = \frac{N\sqrt{h}}{2} S^{ij}}. \quad (29)$$

Appendix N performs the trace and cancellation without suppressing the intermediate ρ -term. Thus

$$\dot{\pi}^{ij} = (\dot{\pi}^{ij})_g + \mathfrak{F}_T^{ij} \quad (30)$$

is shorthand for the spatial-spatial Einstein equation, not an assertion that $\mathfrak{F}_T$ is Hamiltonian. It becomes a Hamiltonian vector-field contribution only after pointwise EMTMatch in a populated $\mathfrak{C}_{\text{src}}^+$ proves

$$\mathfrak{F}_T^{ij} = -\frac{\delta H_m[N, N^i]}{\delta h_{ij}} \quad \text{on the common canonical domain.}$$

This distinction avoids converting a known covariant stress tensor into unconstructed matter phase-space data.

Proposition 21.1 (Ten-component reconstruction). *Equations (25)–(26), the metric kinematics (27), and the forced spatial evolution (30) reconstruct all ten components of (12). The Ward split is the compatibility condition for the source evolution.*

Proof. The scalar and vector constraints set the nn and ni projections of $E_{\mu\nu} = G_{\mu\nu} + \Lambda g_{\mu\nu} - \kappa T_{\mu\nu}$ to zero. Equation (30), after inverting the Legendre map, is (7) and sets the six spatial projections to zero. Lemma 10.1 then gives $E_{\mu\nu} = 0$. Conversely, the covariant equation implies every projection. The contracted Bianchi identity yields $\nabla_\mu T^{\mu\nu} = 0$ for constant κ , which is equivalent to the two Ward equations by Lemma 14.1. $\square$

22. Prequotient reconstruction equivalence

Definition 22.1 (Smooth closed structured source base). Let

$$\widehat{Q}_{\text{sm},0}^{\text{src}} = \left\{ \widehat{x} \in \widehat{Q}^{\text{src}} : T_{\widehat{x}} \text{ is smooth and symmetric, } \mathfrak{C}_\Sigma(T_{\widehat{x}}), J_{\widehat{x}} = 0 \right\}.$$

For each $\widehat{x}$, retain the attachment certificate $\text{Att}_{\widehat{x}}$, complete witness $W_{\widehat{x}}$, and unique representative $T_{\widehat{x}}$. These are source-fibre coordinates, not data to be recovered from $T_{\widehat{x}}$ after they have been discarded.

Definition 22.2 (Source-attached covariant solution family). For $\widehat{x} \in \widehat{Q}_{\text{sm},0}^{\text{src}}$, let $\text{Sol}_{\text{cov}}(\widehat{x}; \mathcal{F})$ consist of tuples

$$(g, \widehat{x}, T_{\widehat{x}}, \text{Att}_{\widehat{x}}, W_{\widehat{x}}, \mathcal{F})$$

satisfying (12) and the declared regularity on the admitted splitting. Define

$$\text{Sol}_{\text{cov}}^{\text{att}} = \coprod_{\widehat{x} \in \widehat{Q}_{\text{sm},0}^{\text{src}}} \text{Sol}_{\text{cov}}(\widehat{x}; \mathcal{F}).$$

Definition 22.3 (Source-attached ADM-history family). For the same $\widehat{x}$, let $\text{Sol}_{\text{ADM}}^{\text{preq}}(\widehat{x})$ consist of histories

$$(h, \pi; N, N^i; \widehat{x}, \text{Att}_{\widehat{x}}, W_{\widehat{x}}; \rho_{\widehat{x}}, j_{\widehat{x}}, S_{\widehat{x}})$$

with h Riemannian and $N \in \mathcal{N}_+$, satisfying the two geometric constraints, (27), (30), the Ward split, and the common regularity conditions, where the projected source triple is required to reconstruct $T_{\hat{x}}$. Define

$$\text{Sol}_{\text{ADM}}^{\text{att,preq}} = \coprod_{\hat{x} \in \widehat{Q}_{\text{sm},0}^{\text{src}}} \text{Sol}_{\text{ADM}}^{\text{preq}}(\hat{x}).$$

The source-attached split map $\mathcal{S}_{\mathcal{F}}^{\text{att}}$ restricts $g, T_{\hat{x}}$, forms (N, N^i, h, K) , applies (17), projects the source, and carries $(\hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}})$ unchanged. The reconstruction map $\mathcal{R}^{\text{att}}$ uses (1) and (9), verifies that the reconstructed tensor is $T_{\hat{x}}$, and returns the retained structured class. It does not claim to infer $\hat{x}$ from a bare tensor.

Theorem W1. 5 (Source-fibre-preserving prequotient reconstruction theorem). On the smooth regular core,

$$\text{Sol}_{\text{cov}}^{\text{att}} \xrightleftharpoons[\mathcal{R}^{\text{att}}]{\mathcal{S}_{\mathcal{F}}^{\text{att}}} \text{Sol}_{\text{ADM}}^{\text{att,preq}}$$

is a fibrewise local reconstructive equivalence modulo the declared foliation-preserving presentation equivalence. Both compositions preserve $\hat{x}$, its attachment certificate, and its complete witness identically.

Proof. Fix $\hat{x}$. Starting from the corresponding covariant tuple, the preceding geometric, source, and canonical sections define the ADM and projected source data without changing the retained fibre coordinate. The projected Einstein and Ward equations place the result in $\text{Sol}_{\text{ADM}}^{\text{preq}}(\hat{x})$. Conversely, an ADM history reconstructs a Lorentzian metric by (1) and a symmetric tensor T' by (9). The defining ancestry condition of the history gives $T' = T_{\hat{x}}$; the map then returns the already retained $(\hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}})$. Proposition 21.1 proves the reconstructed Einstein equation and Lemma 14.1 proves the source ledger.

The algebraic split and reconstruction formulas are inverse wherever $N > 0$, h is nondegenerate, and the declared regularity holds. Fibrewise,

$$\mathcal{R}^{\text{att}} \circ \mathcal{S}_{\mathcal{F}}^{\text{att}} = \text{id}, \quad \mathcal{S}_{\mathcal{F}}^{\text{att}} \circ \mathcal{R}^{\text{att}} = \text{id},$$

modulo the stated presentation equivalence. Taking the disjoint union preserves those identities because neither map changes the fibre label. No global manifold structure on the full solution quotient is asserted. $\square$

Nonclaim 22.4 (No premature refoliation quotient). Theorem W1.5 does not identify histories obtained from different admissible foliations as one physical point. Establishing the full hypersurface-deformation representation, its boundary terms, regularity, characteristic kernel, and quotient faithfulness is the WDW-3 burden.

Part V

Matter canonical lift and boundary extensions

23. Why a tensor source is not a matter Hamiltonian

Module A constructs the first implication

$$\text{attached operational response} \implies T_{\mu\nu} \implies \nabla_{\mu} T^{\mu\nu} = J^{\nu}.$$

It does not construct

$$T_{\mu\nu} \implies (\mathcal{Q}_m, \mathcal{P}_m, \Omega_m, \mathcal{H}_{\perp}^m, \mathcal{H}_i^m).$$

The reverse arrow fails as a general construction principle. A stress tensor may be known phenomenologically or distributionally without a regular Lagrangian. Distinct actions can differ by topological terms, auxiliary fields, improvement structure, or canonical constraint content while agreeing on a restricted

on-shell stress tensor. Singular Legendre maps and internal gauge symmetry introduce primary and Gauss constraints not encoded by the ten stress-tensor components. Thus metric response and canonical continuation carry overlapping but nonidentical information.

Proposition 23.1 (Source/canonical type separation). *No natural map from the full attachment-certified Module-A source image to a unique matter phase space and Hamiltonian constraint package is supplied by WDW-1.*

Proof. Module A admits distributional and non-action-presentable sources, while a canonical phase space requires at least a differentiable configuration domain and a Legendre or equivalent Hamiltonian construction. Even on an action-presentable subdomain, gauge-degenerate and regular actions have different Dirac data. Since those distinctions can change constraint surfaces, observables, and continuation, they are standing-bearing and cannot be removed as skin. Therefore an additional typed certificate is necessary. $\square$

24. Source-canonical-lift certificate

Definition 24.1 (Canonical source family and realization fibres). A canonical matter model $\mathfrak{M}$ carries model-level configuration, phase-space, symplectic, action, Legendre, constraint, Dirac, boundary, and regularity data. Its full matter phase space is denoted $\mathcal{P}_m(\mathfrak{M})$; it is not indexed by one on-shell source class. Let $\text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M})$ denote the declared set of complete smooth canonical solution histories on the regular foliation interval, modulo the foliation-preserving presentation equivalence already used in W1.5. Each such history solves the model’s Hamilton equations and retained constraints, reconstructs smooth covariant matter fields, and lies on the closed-Ward locus.

A *source-realization map* for the model is a well-defined map

$$\text{Src}_{\mathfrak{M}} : \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}) \longrightarrow \hat{\mathcal{Q}}_{\text{sm},0}^{\text{src}}$$

whose value retains the independently admitted Module-A response presentation, attachment certificate, complete witness, and unique tensor representative of the reconstructed covariant history. For $\hat{x}$ in its image, the realization fibre is

$$\text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}; \hat{x}) = \text{Src}_{\mathfrak{M}}^{-1}(\hat{x}).$$

No off-shell phase-space point is thereby promoted to the closed structured-source base.

Certificate 24.2 (Strong canonical source-family lift). For a canonical matter model and a source-realization map, define

$$\mathfrak{C}_{\text{src}}^+(\mathfrak{M}, \text{Src}_{\mathfrak{M}}) = \left(\mathcal{Q}_m, \mathcal{P}_m, \Omega_m, L_m, \mathbb{F}L_m, \mathcal{P}_{\text{cpl}}, \chi_{\mathfrak{M}}, \text{SplitMatch}, \right. \\ \left. \mathcal{H}_{\perp}^m, \mathcal{H}_i^m, \mathcal{D}_{\mathfrak{M}}, \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}), \text{Src}_{\mathfrak{M}}, \text{FibreRet}, \right. \\ \left. \text{EMTMatch}, \text{WardImp}, \text{BoundaryMatch}, \text{Reg}_m \right).$$

Its coordinates require:

1. a matter configuration space, model-level canonical phase space $\mathcal{P}_m(\mathfrak{M})$, and common field domain;
2. a matter symplectic or presymplectic form Ω_m ;
3. a declared Lagrangian or equivalent canonical construction and a Legendre map with its regular, constrained, singular, and stratum status;
4. a coupled model carrier $\mathcal{P}_{\text{cpl}}(\mathfrak{M})$ and an explicit selected-variable map

$$\chi_{\mathfrak{M}} : \mathcal{P}_{\text{cpl}}(\mathfrak{M}) \longrightarrow \mathcal{P}_g^{\infty} \times \mathcal{P}_m(\mathfrak{M});$$

5. proof of the canonical split

$$\Omega_{\text{cpl}} = \chi_{\mathfrak{M}}^*(\Omega_g \oplus \Omega_m)$$

and, after every declared canonical transformation, proof of

$$\mathcal{H}_\perp^{\text{cpl}} = \mathcal{H}_\perp^g + \mathcal{H}_\perp^m, \quad \mathcal{H}_i^{\text{cpl}} = \mathcal{H}_i^g + \mathcal{H}_i^m$$

exactly or modulo named resolved internal constraints and boundary improvements;

6. normal and tangential matter constraint densities on the common smearing domain;
7. a complete Dirac ledger $\mathfrak{D}_\mathfrak{M}$ listing every primary and secondary constraint, its gravitational/deformation/internal type, and one declared classification as first-class, second-class, reducible with its reducibility relation, or explicitly downstream-unresolved; the ledger also records multipliers, bracket domain, boundary improvement, and resolution of every matter-specific second-class sector before downstream eligibility;
8. construction of $\text{Src}_\mathfrak{M}$ on complete canonical solution histories and proof that it is invariant under the declared presentation equivalence;
9. fibre retention: for every history γ , the value $\hat{x}_\gamma = \text{Src}_\mathfrak{M}(\gamma)$ retains $(\text{Att}_{\hat{x}_\gamma}, W_{\hat{x}_\gamma}, T_{\hat{x}_\gamma})$, and no source coordinate is inferred from a bare tensor;
10. pointwise EMTMatch: metric variation along every γ represents exactly $T_{\hat{x}_\gamma}$, with normal, tangential, and spatial projections agreeing with the canonical densities and metric Hamilton equation on that same history;
11. proof that the matter Hamilton equations imply or reproduce the Ward/exchange ledger of $\hat{x}_\gamma$; a converse is reported only if separately proved;
12. matching action and Hamiltonian boundary terms, charges, and fluxes; and
13. one regularity/domain ledger shared by all operations.

Definition 24.3 (Pointed strong lifts). For an admitted structured source class $\hat{x}$, let

$$\mathfrak{L}^+(\hat{x}) = \left\{ \lambda = (\mathfrak{M}, \text{Src}_\mathfrak{M}, \gamma, c_\lambda) : \begin{array}{l} \mathfrak{C}_{\text{src}}^+(\mathfrak{M}, \text{Src}_\mathfrak{M}) \text{ is populated,} \\ \gamma \in \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}), \\ c_\lambda : \text{Src}_\mathfrak{M}(\gamma) = \hat{x} \end{array} \right\}.$$

Thus a point of $\mathfrak{L}^+(\hat{x})$ is not merely a matter model. It is a certified model together with a named complete history lying in the exact realization fibre over $\hat{x}$. Distinct lawful lifts over the same class remain distinct elements of this set.

The certificate is deliberately stronger than action-presentability and stronger than the earlier $\mathfrak{C}_{\text{src}}$ template. An action that produces the right metric derivative may still mix gravitational and matter symplectic variables, fail additive constraint splitting, possess an unanalysed singular Legendre map, omit a primary or reducibility relation, or have a nondifferentiable boundary Hamiltonian, or fail to map its canonical solutions to their exact structured Module-A classes. Such a source remains attached but is not eligible for the canonical handoff.

25. A nonempty source family and pointed occupant: minimally coupled scalar field

Let $\mathfrak{M}_\phi$ denote the minimally coupled real Klein–Gordon canonical model with $V(\phi) = m^2\phi^2/2$. Module A, Proposition 13.3 applies separately to every smooth scalar field with the declared support and boundary behaviour: it constructs that field’s response presentation, attachment certificate, complete smooth witness, and unique tensor representative, and it places every certified Klein–Gordon solution on the closed-Ward locus. Fix the exact scalar solution used by the frozen Einstein–source realization and denote its structured class by

$$\hat{x}_\phi \in \hat{Q}_{\text{sm},0}^{\text{src}}.$$

These upstream data are not manufactured by the ADM split. The model action is

$$S_\phi = -\frac{1}{2} \int_M \sqrt{-g} (g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + m^2 \phi^2) d^4x$$

on the compact boundaryless core. Define the smooth scalar configuration, momentum, and phase-space carriers by

$$\mathcal{Q}_\phi^\infty = C^\infty(\Sigma, \mathbb{R}), \quad \Pi_\phi^\infty = \Gamma^\infty(\text{Dens } \Sigma),$$

$$\mathcal{P}_m(\mathfrak{M}_\phi) = \mathcal{P}_\phi^\infty = \mathcal{Q}_\phi^\infty \times \Pi_\phi^\infty, \quad p_\phi = \frac{\sqrt{h}}{N} (\dot{\phi} - N^i D_i \phi),$$

with the standard smooth Fréchet product topology. The momentum p_ϕ is a weight-one scalar density. On tangent pairs $(\delta\phi_A, \delta p_A)$, define

$$\Omega_\phi((\delta\phi_1, \delta p_1), (\delta\phi_2, \delta p_2)) = \int_\Sigma (\delta p_1 \delta\phi_2 - \delta p_2 \delta\phi_1).$$

Let $\mathcal{A}_{\text{reg}}^\phi = \mathcal{A}_{\text{reg}}(\mathcal{P}_g^\infty \times \mathcal{P}_\phi^\infty)$ be the algebra of smooth functionals with smooth variational derivatives of the corresponding dual tensor and density types, with integrations by parts licensed by compactness. This is the common bracket and Hamiltonian-vector-field domain for the scalar branch.

For $N \in \mathcal{N}_+$ and $N^i \in \mathcal{X}$, the reduced joint Legendre map is

$$\mathbb{F}L_{g\phi}^{\text{red}} : (h, \dot{h}, \phi, \dot{\phi}; N, N^i) \mapsto (h, \pi; \phi, p_\phi),$$

where π is given by (17) and p_ϕ by the preceding display. Its smooth image is

$$\begin{aligned} \mathcal{P}_{\text{cpl}}(\mathfrak{M}_\phi) &= \text{Im } \mathbb{F}L_{g\phi}^{\text{red}} \\ &= \text{Met}^\infty(\Sigma) \times \Gamma^\infty(S^2 T\Sigma \otimes \text{Dens } \Sigma) \times C^\infty(\Sigma, \mathbb{R}) \times \Gamma^\infty(\text{Dens } \Sigma), \end{aligned}$$

because (18) and the scalar velocity inverse below make both velocity blocks regular for $N > 0$. The selected-variable map is the explicit Fréchet diffeomorphism

$$\begin{aligned} \chi_{\mathfrak{M}_\phi} : \mathcal{P}_{\text{cpl}}(\mathfrak{M}_\phi) &\longrightarrow \mathcal{P}_g^\infty \times \mathcal{P}_\phi^\infty, \\ (h, \pi; \phi, p_\phi) &\longmapsto ((h, \pi), (\phi, p_\phi)). \end{aligned}$$

The Legendre map is regular in the scalar velocity for $N > 0$. The matter densities are

$$\mathcal{H}_\perp^\phi = \frac{p_\phi^2}{2\sqrt{h}} + \frac{\sqrt{h}}{2} D_i \phi D^i \phi + \sqrt{h} V(\phi), \quad (31)$$

$$\mathcal{H}_i^\phi = p_\phi D_i \phi. \quad (32)$$

Metric variation gives

$$T_{\mu\nu}^\phi = \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} (\partial_\alpha \phi \partial^\alpha \phi + 2V(\phi)).$$

Its projections obey

$$\sqrt{h} \rho_\phi = \mathcal{H}_\perp^\phi, \quad -\sqrt{h} j_i^\phi = \mathcal{H}_i^\phi.$$

The scalar Hamilton equations reconstruct the Klein–Gordon equation $\square_g \phi - m^2 \phi = 0$, and diffeomorphism covariance gives $\nabla_\mu T_\phi^{\mu\nu} = 0$ on that equation. There is no internal Gauss constraint. On $\mathcal{P}_{\text{cpl}}(\mathfrak{M}_\phi)$, define

$$\Theta_{\text{cpl}} = \int_\Sigma (\pi^{ij} \delta h_{ij} + p_\phi \delta \phi) d^3x, \quad \Omega_{\text{cpl}} = \delta \Theta_{\text{cpl}}.$$

Direct differentiation and regrouping by $\chi_{\mathfrak{M}_\phi}$ give

$$\Omega_{\text{cpl}} = \chi_{\mathfrak{M}_\phi}^*(\Omega_g \oplus \Omega_\phi)$$

and the lapse/shift coefficients are exactly the additive densities $\mathcal{H}_\perp^g + \mathcal{H}_\perp^\phi$ and $\mathcal{H}_i^g + \mathcal{H}_i^\phi$.

Definition 25.1 (Klein–Gordon source-realization map). Let $\text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}_\phi)$ consist of foliation-presentation classes of complete smooth histories

$$\gamma = (h, \pi; \phi, p_\phi; N, N^i)$$

on the regular interval which obey the scalar Hamilton equations (44)–(45), the total deformation constraints, and the retained coupled ADM evolution equations on the frozen branch; have the declared support and endpoint behaviour; and reconstruct a smooth covariant scalar ϕ_γ . Define

$$\begin{aligned} \text{Src}_\phi : \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}_\phi) &\longrightarrow \widehat{Q}_{\text{sm},0}^{\text{src}}, \\ [\gamma] &\longmapsto \widehat{x}[\phi_\gamma] = [\mathfrak{p}_{\phi_\gamma}, \text{Att}_{\phi_\gamma}, W_{\phi_\gamma}]_{\sim_{\text{src}}}, \end{aligned}$$

where the value is the structured class supplied by Module A, Proposition 13.3. Its unique tensor representative is

$$q_g(\text{Src}_\phi[\gamma]) = T^\phi[\phi_\gamma].$$

The map is well-defined because foliation-preserving equivalent histories reconstruct the same covariant scalar and metric, and Module-A attachment and representation are quotient-unique. The Hamilton equations imply the Klein–Gordon equation, so every value lies in the smooth closed source base. For a fixed class, write

$$\text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}_\phi; \widehat{x}) = \text{Src}_\phi^{-1}(\widehat{x}).$$

Definition 25.2 (Named scalar history over the frozen class). Let (g, ϕ) be the exact smooth Einstein–Klein–Gordon realization underlying $\widehat{x}_\phi$, and use the frozen foliation $\mathcal{F}$. Its canonical split is

$$\gamma_\phi = \left[t \longmapsto (h_g(t), \pi_g(t); \phi|_{\Sigma_t}, \sqrt{h_g} \mathcal{L}_{n_g} \phi; N_g(t), N_g^i(t)) \right]_{\mathcal{F}}.$$

The covariant field equation and the split identities give $\gamma_\phi \in \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}_\phi)$. Module-A graph equality and representation uniqueness give

$$\text{Src}_\phi(\gamma_\phi) = \widehat{x}_\phi.$$

Proposition 25.3 (Exact scalar family and pointed strong-lift occupancy). *The pair $(\mathfrak{M}_\phi, \text{Src}_\phi)$ occupies $\mathfrak{C}_{\text{src}}^+(\mathfrak{M}_\phi, \text{Src}_\phi)$ on the compact boundaryless core, and*

$$\lambda_\phi = (\mathfrak{M}_\phi, \text{Src}_\phi, \gamma_\phi, \text{Src}_\phi(\gamma_\phi) = \widehat{x}_\phi) \in \mathfrak{L}^+(\widehat{x}_\phi).$$

In particular, $\text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}_\phi; \widehat{x}_\phi) \neq \emptyset$. The full carrier $\mathcal{P}_\phi^\infty$ belongs to the model $\mathfrak{M}_\phi$; it is not identified with this fixed source fibre.

Proof. The displayed Fréchet carriers, common functional domain, action, symplectic form, regular joint Legendre image, explicit selected-variable diffeomorphism, and additive densities supply the model-level canonical coordinates and SplitMatch. The scalar velocity Hessian is nonsingular. Its complete Dirac ledger contains no scalar primary, Gauss, second-class, or reducibility constraint; the common gravitational primary and deformation candidates carry the explicitly admitted downstream-unresolved status pending G_2/G_3 .

Definition 25.1 supplies the history-to-source map and retains the Module-A attachment certificate, witness, and unique tensor representative for each history. Direct metric variation gives EMTMatch pointwise over this map, and the scalar Hamilton equations imply the corresponding closed Ward ledger. Compactness and absence of a spatial boundary give the trivial spatial BoundaryMatch; fixed scalar endpoint data complete the action variation. Smooth fields populate the common regular domain. Definition 25.2 proves that the exact frozen class has the named occupant γ_ϕ , hence the displayed element of $\mathfrak{L}^+(\widehat{x}_\phi)$. Appendix P verifies each coordinate, including the independent equality $-\delta H_\phi / \delta h_{ij} = N \sqrt{h} S_\phi^{ij} / 2$. $\square$

Nonclaim 25.4 (No class-wide lift). Proposition 25.3 proves model-family and pointed-fibre nonemptiness, not that all Module-A sources are scalar, action-presentable, regular, or canonically equivalent to this model. It also does not identify off-shell scalar phase-space points with closed Module-A source classes.

26. Conditional total constraint package

On a strongly certified source-canonical branch, set

$$\Omega_{\text{tot}} = \Omega_g \oplus \Omega_m, \quad (33)$$

$$\mathcal{H}_{\perp}^{\text{tot}} = \mathcal{H}_{\perp}^g + \mathcal{H}_{\perp}^m, \quad (34)$$

$$\mathcal{H}_i^{\text{tot}} = \mathcal{H}_i^g + \mathcal{H}_i^m. \quad (35)$$

Internal constraints $\mathcal{G}_a$ remain separate:

$$H_{\text{tot}} = \int_{\Sigma} \left(N \mathcal{H}_{\perp}^{\text{tot}} + N^i \mathcal{H}_i^{\text{tot}} + \lambda^a \mathcal{G}_a \right) d^3x + H_{\partial}. \quad (36)$$

They may not be absorbed into $\mathcal{H}_i$ merely because both generate transformations. Their carriers, boundary charges, and quotient kernels can differ.

Definition 26.1 (Conditional canonical domain and constraint map). For a model carrying $\mathfrak{C}_{\text{src}}^+(\mathfrak{M}, \text{Src}\mathfrak{M})$, let $\mathcal{D}_{\text{can}, \mathfrak{M}}$ be the named subset of $\mathcal{P}_g^{\infty} \times \mathcal{P}_m(\mathfrak{M})$ on which every retained constraint density, variational derivative, bracket, and boundary improvement is defined after the declared matter-specific second-class resolution. Let

$$\mathcal{K}_{\mathfrak{M}} = \bigoplus_{\alpha \in I_{\mathfrak{M}}^{\text{unres}}} \Gamma^{\infty}(E_{\alpha, \mathfrak{M}} \otimes \text{Dens } \Sigma)$$

be the direct sum of the unresolved internal-constraint carriers named in $\mathfrak{D}_{\mathfrak{M}}$; it is the zero vector space when that ledger contains no internal constraint. The typed conditional constraint map is

$$\begin{aligned} \Phi_{\text{can}, \mathfrak{M}} : \mathcal{D}_{\text{can}, \mathfrak{M}} &\longrightarrow \Gamma^{\infty}(\text{Dens } \Sigma) \oplus \Gamma^{\infty}(T^*\Sigma \otimes \text{Dens } \Sigma) \oplus \mathcal{K}_{\mathfrak{M}}, \\ z &\longmapsto (\mathcal{H}_{\perp}^{\text{tot}}(z), \mathcal{H}_i^{\text{tot}}(z), \mathcal{G}_{\mathfrak{M}}^{\text{unres}}(z)). \end{aligned}$$

Its prequotient zero locus is $\mathcal{C}_{\text{pre}}^{\text{can}}(\mathfrak{M}) = \Phi_{\text{can}, \mathfrak{M}}^{-1}(0)$. For a pointed lift $\lambda = (\mathfrak{M}, \text{Src}\mathfrak{M}, \gamma, c_{\lambda})$, the notation $\mathcal{D}_{\text{can}, \lambda}$, $\mathcal{K}_{\lambda}$, $\Phi_{\text{can}, \lambda}$, and $\mathcal{C}_{\text{pre}}^{\text{can}}(\lambda)$ means these model-level objects together with the retained realization map, fibre, and distinguished history. For the scalar model,

$$\mathcal{D}_{\text{can}, \mathfrak{M}_{\phi}} = \mathcal{P}_g^{\infty} \times \mathcal{P}_{\phi}^{\infty}, \quad \mathcal{K}_{\mathfrak{M}_{\phi}} = 0$$

on the smooth compact core, and its functional bracket domain is $\mathcal{A}_{\text{reg}}^{\phi}$.

Theorem W1. 6 (Strongly conditional total-source canonical theorem). For an attachment-certified source class $\hat{x}$, a source-faithful total canonical handoff exists only relative to a pointed lift $\lambda \in \mathfrak{L}^+(\hat{x})$. On that lift the coupled symplectic carrier is canonically identified with the displayed gravitational–matter split, the deformation constraints are additive in the selected variables, and the complete Dirac ledger is retained. The full phase space belongs to the model in λ ; source faithfulness is supplied by its realization map, exact fibre, and distinguished history, not by identifying that phase space with $\hat{x}$. Every package reported as such must supply those certificate coordinates. When $\mathfrak{L}^+(\hat{x}) = \emptyset$, WDW-1 reports only the source-attached covariant projections and gravitational forced-history response.

Proof. Given $\lambda = (\mathfrak{M}, \text{Src}\mathfrak{M}, \gamma, c_{\lambda}) \in \mathfrak{L}^+(\hat{x})$, SplitMatch proves rather than assumes the model-level direct-sum symplectic form and the sums (34)–(35) on the common domain. The complete Dirac ledger retains all internal constraints, their class, reducibility, multipliers, boundary improvements, and second-class resolution. FibreRet carries $(\hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}}, T_{\hat{x}})$ through $\text{Src}\mathfrak{M}(\gamma) = \hat{x}$. Pointwise EMTMatch gives $\mathcal{H}_{\perp}^m = \sqrt{\hbar} \rho$ and $\mathcal{H}_i^m = -\sqrt{\hbar} j_i$ in the frozen convention; hence Theorem 20.1 identifies the total canonical and geometric constraints. WardImp proves that the matter Hamilton equations imply or reproduce (10)–(11); no converse is inferred without a separate proof. BoundaryMatch makes (36) differentiable and preserves the source flux ledger.

Conversely, a source-faithful total package with the asserted exact matches necessarily has an on-shell realization map and exact source fibre in addition to matter variables and momenta, a coupled and

split (pre)symplectic comparison, a Hamiltonian construction and Legendre status, additive or precisely corrected normal and tangential constraints, a complete Dirac ledger, the matching proofs, and a common domain. These are exactly the strong family-certificate and pointed-lift coordinates. If any is absent, the canonical handoff is not licensed, although the source-attached split and reconstruction of Theorems W1.2 and W1.5 remain available. $\square$

27. Compact boundaryless core

For $\partial\Sigma = \emptyset$, integrations by parts used in deriving $\mathcal{H}_i^g$, varying the curvature term, and checking differentiability have no spatial surface contribution. With the declared smooth variations,

$$H_{\partial} = 0, \quad Q_{\partial} = 0, \quad \text{Flux}_{\partial\Sigma} = 0.$$

This statement concerns the spatial boundary. Initial and final Cauchy surfaces remain part of the variational endpoint ledger.

Proposition 27.1 (Core differentiability). *The gravitational generator $H_g[N, N^i]$ is differentiable on the smooth compact boundaryless core for smooth lapse and shift, and lapse and shift enter only as bulk multipliers.*

Proof. Variation of the curvature term produces a spatial divergence, variation of the momentum constraint produces a spatial divergence, and both integrate to zero because $\partial\Sigma = \emptyset$. The remaining bulk terms define the functional derivatives used in (27)–(28). Since neither $\dot{N}$ nor $\dot{N}^i$ occurs, their multiplier status follows from Theorem W1.4. $\square$

28. Boundary and asymptotic extensions

Certificate 28.1 (Boundary differentiability).

$$\mathfrak{C}_{\partial} = (\partial M, \mathcal{B}, S_{\partial}, \delta S_{\partial}, H_{\partial}, Q_{\partial}, \text{Flux}_{\partial}, \text{TimeStatus}_{\partial}).$$

It specifies the boundary causal geometry, allowed variations, action completion, Hamiltonian surface term, charge, source flux, fixed or variable lapse/shift boundary values, and whether a nonzero boundary Hamiltonian generates physical evolution.

For a finite boundary one must state boundary conditions and a surface generator. For an asymptotically flat branch one additionally needs falloff and parity conditions; the ADM energy and momentum can then be nonzero [20]. An AdS-like branch needs its own boundary conditions and counterterm or equivalent charge prescription. Null boundaries require a different action and charge formalism. None is silently merged into the compact core.

Branch	Required added data	Temporal-data status
Finite spatial boundary	boundary conditions, surface term, source flux, differentiable generator	standing-bearing if paired with nonzero generator or charge
Asymptotically flat	falloff, parity, ADM charges	asymptotic time translation may be physical
AdS-like	boundary conditions, counterterms or equivalent charge	boundary time is a declared anchor/tensor
Null boundary	null action, corner terms, null charges	separate target update
Open/effective source	named J^ν , external carrier, flux balance	not the closed-source core

29. Lapse, shift, and temporal standing

Three distinct facts must not be conflated:

1. t labels the chosen foliation.
2. N, N^i are bulk multipliers on the compact core.
3. A boundary value of N or N^i can pair with a nonzero differentiable charge.

The first does not make t an operational clock. The second does not prove all bulk refoliations are already the physical characteristic kernel. The third prevents charge-bearing boundary data from being discarded as skin.

Theorem W1. 7 (Boundary standing and lapse/shift classification). On the compact boundaryless core, lapse and shift are differentiable bulk multipliers and the foliation label is not thereby a physical clock. On any extension carrying $\mathfrak{C}_\partial$, boundary lapse/shift or asymptotic temporal data are standing-bearing whenever variation changes a nonzero differentiable generator, charge, flux, or continuation. An external parameter absent from the state, boundary, background, and readout ledgers cannot acquire same-scope clock standing.

Proof. The first clause is Proposition 27.1 together with the no-external-clock import theorem: occurrence as a presentation parameter is not a clock-standing certificate. For the second, $\mathfrak{C}_\partial$ makes the variation of the total Hamiltonian well defined. If a boundary datum pairs with nonzero Q_∂ or changes flux or Hamiltonian continuation, varying it changes standing-bearing output and it is therefore anchor/tensor rather than skin. The final clause is the imported fixed-scope clock theorem. None of these steps identifies the complete bulk refoliation kernel, which remains WDW-3 work. $\square$

Part VI

AASC role typing and exact WDW handoff

30. Anchor–tensor–skin ledger

Role typing is relative to the frozen comparison class. An item is an anchor when changing it changes the target envelope; a tensor when it changes invariant content, admissible continuation, or readout; and skin when it changes neither. “Tensor” here is an AASC role word and need not mean a spacetime tensor field.

Table 3: WDW-1 anchor–tensor–skin ledger.

Object	WDW-1 role	Qualification
Dimension and Lorentz signature	Anchor	Frozen projection envelope.
Global hyperbolicity and Cauchy topology	Anchor	Not derived from the four kernel roles in this paper.
$g_{\mu\nu}$	Standing-bearing projection carrier/tensor	Downstream of the AASC gravity role.
∇^{LC}	Derived connection occupant	Unique only in $\text{MinMetricNat}_1(M, g)$.
h_{ij}, K_{ij}, π^{ij}	Canonical tensor/presentation data	Standing-bearing through the reconstruction profile.

Object	WDW-1 role	Qualification
Coordinate labels and chart components	Skin	Only when reconstructed content and continuation are unchanged.
Lapse and shift in compact bulk	Multiplier/presentation	Complete characteristic verdict awaits WDW-3.
Boundary lapse/shift with nonzero charge	Anchor/tensor	Cannot be quotiented as bulk skin.
ρ, j_i, S_{ij}	Source projections	Must descend from one Module-A class.
Matter Hamiltonian densities	Tensor/continuation load	Require a populated source-family certificate and pointed lift.
Choice of foliation	Presentation candidate	Physical neutrality across refoliations awaits WDW-3.
κ, Λ	Projection/calibration data	Values are not kernel-generated.
H_∂, Q_∂	Boundary tensor/anchor	Standing-bearing when nonzero or required for differentiability.
Ordering, regulator, domain, measure	Outside WDW-1	Later tensor/skin classification depends on physical effect.

31. AER comparison and field-theoretic lift boundary

The imported regular AER theorem is finite-dimensional/local. WDW-1 constructs formal role occupants resembling its state carrier, standing surface, multipliers, and constraint generators, but the infinite-dimensional promotion is a separate certificate:

Certificate 31.1 (Field-theoretic AER lift).

$$\mathfrak{C}_{\text{AER}}^{\text{field}} = (\mathcal{P}^s, \Omega^s, \Phi, \mathcal{C}, \pi_{\text{phys}}, \text{RankReg}, \text{Char}, \text{ObsSep}, \text{ObsMax}, \text{Domain}, \text{Boundary}).$$

It requires:

1. a specified Sobolev/Banach, tame Fréchet, or other controlled completion;
2. a well-defined symplectic or presymplectic structure on a common domain;
3. a smooth constraint map Φ and constraint surface $\mathcal{C}$;
4. a constant-rank regular stratum or an explicit stratified substitute;
5. defined Hamiltonian vector fields and their domains;
6. a faithful quotient, groupoid, or equivalent descent object;
7. observable separation and maximality;
8. identification of the characteristic distribution;
9. closure of spatial, internal, and normal-deformation generators; and
10. boundary compatibility retaining all nonzero charges.

Proposition 31.2 (No finite-to-field type jump). *The local finite-dimensional AER forcing theorem does not by itself establish $\mathfrak{C}_{\text{AER}}^{\text{field}}$ for the ADM field phase space.*

Proof. In infinite dimensions, existence of Hamiltonian vector fields, closed range, constant rank, smooth quotient structure, and even nondegeneracy properties of a two-form depend on the chosen completion and domains. The imported theorem assumes the corresponding regular local objects. WDW-1 supplies

candidate formulas but not those analytic conclusions. Therefore applying the finite theorem without Certificate 31.1 would omit hypotheses. $\square$

WDW-2 receives the spatial-diffeomorphism and internal-gauge coordinates of this certificate. WDW-3 receives the normal-deformation, characteristic, and refoliation coordinates. The division prevents either later paper from using the other’s conclusion as an unproved premise.

32. Coordinate time, physical time, and standing

The following objects are distinct:

Foliation label t . A coordinate on the selected splitting.

Proper time. Metric length along a timelike worldline; it depends on g and the worldline.

Lapse N . The conversion between coordinate separation and normal proper-time separation in the chosen split.

Boundary time. A parameter associated with boundary conditions and potentially a nonzero boundary Hamiltonian.

Internal clock. A subsystem carrying a proved clock-standing certificate and relational readout.

Standing order. The non-temporal upstream order induced by admissible irreversible continuation.

External Schrödinger parameter. A parameter requiring an independent same-scope authorization in a closed constrained system.

Proposition 32.1 (Presentation time is not clock standing). *A foliation label does not acquire physical-clock standing solely by appearing in (1), and bulk multiplier status does not imply absence of relational change.*

Proof. Clock standing requires a target, comparison profile, readout, admissible continuation, and quotient-faithful use. The coordinate label supplies none of those by itself. Conversely, relational change concerns correlations among standing-bearing fields or subsystems and can be nontrivial even when the bulk Hamiltonian is a sum of constraints. Boundary time and internal clocks remain admissible under their own certificates. $\square$

33. WDW-1 capstone theorem

Definition 33.1 (Source-index partition). Let

$$I_{\text{src}} = \left\{ \hat{x} \in \hat{Q}_{\text{sm},0}^{\text{src}} : \hat{x} \text{ satisfies the attachment, closed-Ward, and Cauchy hypotheses of W1.8} \right\}.$$

With $\hat{x}_0$ the attached zero-witness class of Module A, Example 13.2, define

$$\begin{aligned} I_{\text{vac}} &= \{\hat{x}_0\}, \\ I_{\text{can}} &= \left\{ \hat{x} \in I_{\text{src}} \setminus I_{\text{vac}} : \mathfrak{L}^+(\hat{x}) \neq \emptyset \right\}, \\ I_{\text{forced}} &= \left\{ \hat{x} \in I_{\text{src}} \setminus I_{\text{vac}} : \mathfrak{L}^+(\hat{x}) = \emptyset \right\}. \end{aligned}$$

The vacuum normalization has precedence: even if an equivalent trivial canonical presentation is later supplied for $\hat{x}_0$, its WDW-1 handoff remains the vacuum handoff.

Proposition 33.2 (Disjointness and exhaustiveness). *The three index sets in Definition 33.1 are pairwise disjoint and satisfy*

$$I_{\text{src}} = I_{\text{vac}} \amalg I_{\text{can}} \amalg I_{\text{forced}}.$$

Consequently, every admitted structured source ancestry receives exactly one WDW-1 handoff type and one downstream eligibility value.

Proof. The last two sets exclude I_{vac} , so neither meets the vacuum set. They are mutually disjoint because nonemptiness and emptiness of the same pointed-lift set are complementary predicates on the fixed source class. Every nonvacuum element of I_{src} lies in exactly one of those two sets. Adding the unique normalized vacuum class proves the displayed partition and the uniqueness claim. $\square$

Definition 33.3 (Vacuum handoff). Let $\hat{x}_0$ denote the attached smooth zero-witness class of Module A, Example 13.2. The vacuum handoff is

$$\mathfrak{H}_{\text{W1}}^{\text{vac}} = (\hat{x}_0, \mathcal{P}_g^\infty, \Omega_g, \Phi_g, \mathcal{C}_{\text{vac}}, \mathcal{N}, \mathcal{N}_+, \mathcal{X}, \text{Recon}_{\hat{x}_0}, \mathfrak{E}_\nabla, \mathfrak{E}_\partial).$$

Here Φ_g , $\mathcal{C}_{\text{vac}}$, and all smearing and functional domains are Definitions 18.3–18.2. The source label is retained to distinguish an upstream-certified zero response from an arbitrary zero assignment. This handoff is eligible for WDW-2 and WDW-3.

Definition 33.4 (Strongly certified canonical handoff). For $\hat{x} \in I_{\text{can}}$ and

$$\lambda = (\mathfrak{M}, \text{Src}_{\mathfrak{M}}, \gamma, c_\lambda) \in \mathfrak{L}^+(\hat{x}),$$

define

$$\mathcal{P}_{\text{can}}(\lambda) = \mathcal{P}_g^\infty \times \mathcal{P}_m(\mathfrak{M}), \quad \Omega_{\text{can}}(\lambda) = \Omega_g \oplus \Omega_m.$$

The domain $\mathcal{D}_{\text{can},\lambda}$, internal carrier $\mathcal{K}_\lambda$, typed map $\Phi_{\text{can},\lambda}$, and zero locus $\mathcal{C}_{\text{pre}}^{\text{can}}(\lambda)$ are those of Definition 26.1; every retained internal constraint and its reducibility data occur in $\mathfrak{D}_{\mathfrak{M}}$. The source-faithful component of the handoff is

$$(\text{Src}_{\mathfrak{M}}, \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}; \hat{x}), \gamma, c_\lambda);$$

the full phase space remains model-level and is not asserted to be this solution fibre. Then

$$\mathfrak{H}_{\text{W1}}^{\text{can}}(\lambda) = \left(\begin{array}{c} \hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}}, \mathfrak{M}, \text{Src}_{\mathfrak{M}}, \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}; \hat{x}), \gamma, c_\lambda, \\ \mathcal{P}_{\text{can}}(\lambda), \Omega_{\text{can}}(\lambda), \\ \mathcal{D}_{\text{can},\lambda}, \mathcal{K}_\lambda, \Phi_{\text{can},\lambda}, \mathcal{C}_{\text{pre}}^{\text{can}}(\lambda), \\ \mathcal{N}, \mathcal{N}_+, \mathcal{X}, \mathfrak{D}_{\mathfrak{M}}, \text{Recon}_{\hat{x}}, \mathfrak{E}_\nabla, \mathfrak{E}_\partial \end{array} \right).$$

It is eligible for WDW-2/WDW-3 subject to their still-open quotient and deformation-algebra proofs. If several lawful lifts lie over $\hat{x}$, none is selected or collapsed:

$$\mathfrak{H}_{\text{W1}}^{\text{can}}[\hat{x}] = \coprod_{\lambda \in \mathfrak{L}^+(\hat{x})} \mathfrak{H}_{\text{W1}}^{\text{can}}(\lambda).$$

Eligibility does not assert that the downstream proofs already hold.

Definition 33.5 (Attached noncanonical forced handoff). For $\hat{x} \in I_{\text{forced}}$, let $\text{Sol}_{\text{ADM}}^{\text{att,forced}}(\hat{x})$ be its fibre in Definition 22.3. Define the geometric relation along a history by

$$\mathfrak{E}_x(t) = \left(\mathcal{H}_\perp^g + \sqrt{h} \rho_{\hat{x}}, \mathcal{H}_i^g - \sqrt{h} j_{i,\hat{x}} \right)$$

and retain the explicitly calculated forcing $\mathfrak{F}_{T_x}^{ij} = N\sqrt{h} S_x^{ij}/2$. The forced handoff is

$$\mathfrak{H}_{\text{W1}}^{\text{forced}}(\hat{x}) = (\text{Sol}_{\text{ADM}}^{\text{att,forced}}(\hat{x}), \mathcal{P}_g^\infty, \Omega_g, \mathfrak{E}_x, \mathfrak{F}_{T_x}, \text{Recon}_{\hat{x}}, \hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}}, \mathfrak{E}_\nabla, \mathfrak{E}_\partial).$$

The relation $\mathfrak{E}_x(t) = 0$ is not a Hamiltonian moment map on a matter phase space. This handoff is blocked from constraint-algebra and quotient work until a strong source-family certificate and pointed lift are supplied.

Definition 33.6 (Typed WDW-1 handoff family and eligibility). The WDW-1 output is the disjoint typed family

$$\mathfrak{H}_{\text{W1}} = \mathfrak{H}_{\text{W1}}^{\text{vac}} \amalg \coprod_{\hat{x} \in I_{\text{can}}} \coprod_{\lambda \in \mathfrak{L}^+(\hat{x})} \mathfrak{H}_{\text{W1}}^{\text{can}}(\lambda) \amalg \coprod_{\hat{x} \in I_{\text{forced}}} \mathfrak{H}_{\text{W1}}^{\text{forced}}(\hat{x}),$$

together with

$$\text{Elig}_{23}(\mathfrak{H}) = \begin{cases} \text{ELIGIBLE}, & \mathfrak{H} = \mathfrak{H}_{\text{W1}}^{\text{vac}} \text{ or } \mathfrak{H}_{\text{W1}}^{\text{can}}(\lambda) \text{ with } \lambda \in \mathfrak{L}^+(\hat{x}), \\ \text{BLOCKED}, & \mathfrak{H} = \mathfrak{H}_{\text{W1}}^{\text{forced}}(\hat{x}). \end{cases}$$

Elig_{23} is single-valued on source ancestry by Proposition 33.2: every retained lift over a canonical source class is eligible, and no such class occurs in the forced branch. $\mathfrak{C}_\partial$ is trivial on the compact core and must be populated on every boundary extension.

Theorem W1. 8 (WDW-1 canonical projection endpoint). Let $\mathcal{T} \in \mathcal{T}_{\text{W1}}^{0,\nabla}$ satisfy:

1. target adequacy, kernel faithfulness, and the accepted realized-physical-interior dependency chain;
2. the accepted AER gravity-role and Einstein minimal metric-projection packages;
3. occupation of $\text{MinMetricNat}_1(M, g)$ with certified smooth standing-preserving finite transport;
4. global hyperbolicity and the declared regular splitting;
5. an attachment-certified smooth Cauchy-admissible closed Module-A source; and
6. the downstream Einstein–Hilbert/ADM representation on the common regular domain.

Then:

1. $\nabla = \nabla^{\text{LC}}$;
2. (ρ, j, S) descend from one retained structured source fibre $(\hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}}, T_{\hat{x}})$;
3. the normal and tangential Einstein projections give the ADM geometric constraints;
4. the gravitational Legendre transform gives (h, π) and multiplier variables (N, N^i) ;
5. the constraint-plus-evolution system and the covariant Einstein–source system are locally source-fibre-preservingly reconstructively equivalent at the prequotient level;
6. a total matter-coupled canonical package over a structured source class exists only relative to a pointed strong lift in $\mathfrak{L}^+(\hat{x})$, whose model-level phase space, source-realization map, exact fibre, and distinguished history are all retained;
7. the compact core has no spatial boundary Hamiltonian, while every boundary extension retains its differentiability, charge, flux, and temporal-status data; and
8. the typed handoff family $\mathfrak{H}_{\text{W1}}$ and its downstream eligibility map are constructed on a disjoint exhaustive source-index partition without identifying its three branch types or collapsing multiple lawful canonical lifts over one source class.

This closes the canonical-projection gate G_1 by constructing and classifying every source branch. Only eligible branches may progress. It does not close the spatial/internal proper-redescription quotient G_2 , the full refoliation/deformation gate G_3 , a class-wide matter-occupancy gate G_4 , or any quantum gate.

Proof. Theorem W1.1 gives the dependency and connection conclusion on $\mathcal{T}_{\text{W1}}^{0,\nabla}$. Theorem W1.2 gives source descent and Ward equivalence. Theorem W1.3 gives the geometric constraints. Theorem W1.4 gives the gravitational Legendre and multiplier structure, and Theorem 20.1 identifies the geometric and canonical forms. Theorem W1.5 gives local prequotient reconstruction. Theorem W1.6 proves the exact conditional matter statement. Theorem W1.7 and Proposition 27.1 give the compact and conditional boundary classification. Definitions 18.1–18.3 construct the gravitational carrier, two-form, domains, constraint map, and vacuum locus. Certificate 24.2, Definition 24.3, and Theorem W1.6 construct a canonical branch only when the source class has a populated pointed lift. The history-to-source map and its exact fibre prevent the model phase space from being misidentified with one source instance. Definition 33.5 retains noncanonical sources as forced histories without promoting their geometric relations to a moment map. Proposition 33.2 proves that the vacuum normalization and the two nonvacuum pointed-lift alternatives classify every admitted source ancestry exactly once. Collecting the

three disjoint objects gives Definition 33.6. The nonclaims follow from Proposition 31.2, Nonclaim 22.4, and the absence of any quantum operator construction. $\square$

34. Exact handoff to WDW-2 and WDW-3

WDW-2 receives only a handoff with $\text{Elig}_{23} = \text{ELIGIBLE}$. For the vacuum branch it receives:

1. $(\mathcal{P}_g^\infty, \Omega_g, \Phi_g, \mathcal{C}_{\text{vac}})$;
2. the spatial-diffeomorphism generator candidate on $\mathcal{A}_{\text{reg}}(\mathcal{P}_g^\infty)$;
3. the smearing spaces $\mathcal{N}, \mathcal{X}$ and positive reconstruction lapse $\mathcal{N}_+$;
4. the boundary charge ledger; and
5. the attached zero-source, connection, and reconstruction ancestry.

For a strongly certified canonical branch with $\lambda = (\mathfrak{M}, \text{Src}_{\mathfrak{M}}, \gamma, c_\lambda) \in \mathfrak{L}^+(\hat{x})$, it instead receives

$$\begin{aligned} & \mathcal{P}_{\text{can}}(\lambda), \Omega_{\text{can}}(\lambda), \\ & (\mathcal{D}_{\text{can}, \lambda}, \mathcal{K}_\lambda, \Phi_{\text{can}, \lambda}, \mathcal{C}_{\text{pre}}^{\text{can}}(\lambda),) \\ & \text{Src}_{\mathfrak{M}}, \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}; \hat{x}), \gamma, c_\lambda \end{aligned}$$

and the complete Dirac ledger, including every unresolved internal generator candidate. Every lawful λ is retained; WDW-2 may not silently choose among multiple lifts over the same source class. Its burden is to prove the proper spatial/internal quotient while retaining improper boundary symmetries and charges. A forced branch receives no WDW-2 generator assignment.

WDW-3 likewise receives only an eligible vacuum or strongly certified canonical handoff:

1. the Hamiltonian-constraint candidate and lapse deformation family;
2. hypersurface embeddings and the reconstruction map;
3. boundary deformation data;
4. the unresolved characteristic/refoliation status; and
5. the remaining coordinates of $\mathfrak{C}_{\text{AER}}^{\text{field}}$.

Its burden is the hypersurface-deformation algebra, anomaly-free classical closure on the declared domain, faithful characteristic identification, and refoliation descent.

Stop condition. A noncanonical forced handoff cannot enter WDW-2 or WDW-3. No quantum gravitational constraint is defined until WDW-2 and WDW-3 close their respective quotient and deformation gates on one common eligible target.

Part VII

Adversarial closure and publication standard

35. Adversarial map

Every objection must be tested against the exact theorem chain before a verdict. An objection already answered by a named certificate survives only if it identifies a false hypothesis, invalid inference, or failed instantiation inside that certificate.

Table 4: Adversarial verification map.

Likely attack	Required theorem-level response
“This is standard ADM with AASC labels.”	The ADM identities are not claimed as novel. Test the dependency theorem, connection ancestry, source attachment ancestry, canonical-lift gate, boundary-standing theorem, and exact handoff.
“Levi–Civita was assumed.”	Inspect $\mathfrak{C}_\nabla$, Proposition 7.2, and Theorem W1.1.
“The foliation selected the connection.”	Theorem W1.1 closes before Section 8 introduces any foliation.
“Global hyperbolicity was derived from AASC.”	Denied; it is an explicit anchor of $\mathcal{T}_{W1}^0$.
“The source tensor was attached arbitrarily.”	Test Module-A graph equality, structured quotient, representation uniqueness, and Table 1.
“Every Module-A source was given ADM matter data.”	Denied; the smooth Cauchy certificate and the source-family/pointed-lift gate are separate.
“Every distribution restricts to Σ .”	Denied; the core is smooth and Appendix H records the microlocal obstruction.
“Stress-energy determines a unique matter Hamiltonian.”	Denied by Proposition 23.1; only certified lifts are reported.
“The reconstruction forgets the structured source.”	Denied by the fibrewise disjoint-union construction in Theorem W1.5: both maps carry $\hat{x}$, $\text{Att}_{\hat{x}}$, and $W_{\hat{x}}$ unchanged.
“One phase space silently covers every source branch.”	Denied by Definitions 33.3–33.5; vacuum, strongly canonical, and forced histories have distinct carriers and eligibility.
“The vacuum source also enters the blocked forced coproduct.”	Denied by Definition 33.1 and Proposition 33.2; the vacuum normalization is removed from both nonvacuum index sets, which are complementary on the remaining source classes.
“One fixed scalar source class labels the full Klein–Gordon phase space.”	Denied by Definitions 24.1 and 25.1. The full phase space belongs to $\mathfrak{M}_\phi$; complete histories map to their own structured classes, and $\hat{x}_\phi$ is the exact fibre containing γ_ϕ . Off-shell points are not promoted to the closed source base.
“Several canonical realizations over one source class were silently identified.”	Denied by Definition 33.4; the handoff above $\hat{x}$ is the coproduct over every $\lambda \in \mathfrak{L}^+(\hat{x})$.
“The canonical certificate assumes a direct sum.”	Denied; SplitMatch must prove the coupled-to-split symplectic comparison and additive constraints, and $\mathfrak{D}_{\mathfrak{M}}$ retains the complete Dirac ledger.
“The source forcing was defined to match matter variation.”	Denied by (29) and Appendix N; the term is derived from the spatial Einstein equation and independently reobtained on the scalar branch.
“The Hamiltonian constraint was assumed.”	Its geometric antecedent is Theorem W1.3; its canonical expression is derived by the Legendre map and Theorem 20.1.
“Lapse and shift were declared gauge.”	WDW-1 proves multiplier status and a boundary-sensitive classification; full characteristic status is deferred.

Likely attack	Required theorem-level response
“AER already proves the ADM quotient.”	Denied by Proposition 31.2 and Certificate 31.1.
“Source conservation was assumed from GR.”	It is imported from the closed Module-A Ward ledger and independently recombined in Lemma 14.1; consistency with Bianchi is then checked.
“The action reverses dependency.”	Proposition 16.1 treats it as a downstream equivalent representation.
“Boundary terms are optional.”	Denied; nonzero or differentiability-required terms are retained by $\mathfrak{C}_\partial$.
“External time is impossible.”	Not claimed. Undeclared same-scope import is blocked; boundary, relational, and effective time remain available when certified.
“WDW-1 proves the frozen formalism.”	Denied; it constructs only the classical canonical antecedent.
“The paper derives the Wheeler–DeWitt equation.”	Denied; no functional derivative or quantum operator occurs as a theorem step.
“Torsion or Palatini gravity is excluded absolutely.”	Denied; those are changed connection targets, not occupants of the minimal branch.
“The result solves all canonical gravity.”	Denied; singular strata, global quotient geometry, all boundaries, and all matter sectors are outside the core.
“The scalar example proves universal matter occupancy.”	Denied explicitly after Proposition 25.3.

36. Failure conditions and falsifiers

The WDW-1 endpoint fails, rather than merely becoming less elegant, if any of the following is shown on the frozen target:

1. the Einstein target does not occupy the minimal metric branch;
2. the connection theorem is imported outside its finite-rank hypotheses;
3. the smooth global split or declared regularity hypotheses fail;
4. the source lacks a Cauchy projection on the claimed branch;
5. (ρ, j, S) does not descend from the same Module-A representative;
6. a Gauss–Codazzi sign or term is inconsistent with the convention ledger;
7. the Legendre map, Hamiltonian density, or multiplier calculation is invalid;
8. constraints plus spatial evolution fail to reconstruct the covariant equation;
9. a matter Hamiltonian is asserted without a populated source-family certificate and pointed lift;
10. a model-level matter phase space is identified with one fixed structured source class, or no canonical-history-to-source realization map and exact fibre are retained;
11. the structured source coordinate, attachment certificate, or witness is lost under either reconstruction map;
12. the vacuum, certified-canonical, and forced-source handoffs are assigned one untyped phase space or constraint surface, or one source ancestry receives two incompatible handoff types;
13. a boundary generator is nondifferentiable or a nonzero charge/flux is omitted;

14. characteristic or refoliation closure is reported without $\mathfrak{C}_{\text{AER}}^{\text{field}}$; or
15. a quantum claim is used to close a classical gate.

37. Scope and reportability

Reportable	Not reportable
Connection ancestry on the minimal branch	Unique connection across richer targets
Attachment-safe smooth Cauchy source split	Arbitrary distributional trace
ADM geometric constraint projection	Class-wide canonical matter existence
Gravitational Legendre and multiplier structure	Complete reduced field phase space
Source-fibre-preserving local prequotient reconstruction	All refoliations are gauge in every boundary regime
Strongly conditional total source-canonical lift	Full Standard Model or fermionic coupling
Boundary standing criterion	All asymptotic or null boundary closures
Typed WDW-2/3 handoff with eligibility map	WDW operator, anomalies, physical inner product, or quantum dynamics

38. Conclusion

WDW-1 does not quantize the accepted Einstein endpoint. It constructs the canonical gravitational presentation that a later constraint-native quantum realization must faithfully occupy, while preserving the ancestry of its connection, source, boundary, and temporal classifications. On the frozen smooth compact target, the result is complete at the prequotient classical level: Levi–Civita closure precedes foliation on the populated connection subtype; the source split and inverse maps preserve one complete structured source fibre; the Einstein projections give the geometric constraints; the Einstein–Hilbert representation gives the gravitational Legendre and multiplier structure; and constraints plus evolution reconstruct the covariant equation.

The limitations are structural results, not omissions to be hidden. A stress tensor is not a matter Hamiltonian; a tensor distribution has no automatic Cauchy trace; a local finite-dimensional AER theorem is not a global field-theoretic quotient; multiplier status is not yet complete refoliation closure; and coordinate time is not an operational clock merely because it labels an ADM split. By exposing those boundaries, the manuscript closes exactly one gate and hands later papers a typed family rather than an ambiguous formal equation: the normalized vacuum and strongly certified nonvacuum canonical branches may proceed, while an attached noncanonical nonvacuum source is stopped at the forced-history interface. The three source-index sets are disjoint and exhaustive, so no source ancestry receives contrary eligibility reports. On the canonical branch, the full phase space is model-level: source faithfulness is carried by an on-shell realization map, its exact fibres, and a distinguished history, with every lawful lift retained rather than silently selected.

A. Convention and sign ledger

Table 5: Frozen convention ledger.

Item	Convention	Immediate consequence
Signature	$(- + ++)$	$n^\mu n_\mu = -1.$
Riemann tensor	$[\nabla_\mu, \nabla_\nu]v^\rho = R^\rho_{\sigma\mu\nu}v^\sigma$	$R_{\sigma\nu} = R^\rho_{\sigma\rho\nu}.$
Spatial projector	$h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$	$g_{\mu\nu} = h_{\mu\nu} - n_\mu n_\nu.$

Item	Convention	Immediate consequence
Extrinsic curvature	$K_{ij} = h_i^\mu h_j^\nu \nabla_\mu n_\nu = \frac{1}{2} \mathcal{L}_n h_{ij}$	$\dot{h}_{ij} = 2N K_{ij} + \mathcal{L}_{\vec{N}} h_{ij}$.
Acceleration	$a_i = D_i \ln N$	$\nabla_\mu n_\nu = K_{\mu\nu} - n_\mu a_\nu$.
Einstein equation	$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$	$G_{nn} - \Lambda = \kappa \rho$.
Gravitational action	$\frac{1}{2\kappa} \int \sqrt{-g} (R - 2\Lambda) + S_\partial$	$\mathcal{L}_g = \frac{N\sqrt{h}}{2\kappa} (R^{(3)} + K_{ij} K^{ij} - K^2 - 2\Lambda)$.
Canonical momentum	$\pi^{ij} = \frac{\sqrt{h}}{2\kappa} (K^{ij} - K h^{ij})$	$K_{ij} = \frac{2\kappa}{\sqrt{h}} (\pi_{ij} - \frac{1}{2} \pi h_{ij})$.
Energy density	$\rho = T_{\mu\nu} n^\mu n^\nu$	$\mathcal{H}_\perp^m = \sqrt{h} \rho$ on a certified lift.
Momentum density	$j_i = -h_i^\mu n^\nu T_{\mu\nu}$	$\mathcal{H}_i^m = -\sqrt{h} j_i$ on a certified lift.
Spatial stress	$S_{ij} = h_i^\mu h_j^\nu T_{\mu\nu}$	$S = h^{ij} S_{ij}$.
Boundary orientation	outward spatial normal and future/past endpoint normals	Signs of GHY and Hamiltonian charges follow the selected branch.

A.1 Independent sign checks

Three compact tests detect the most common integration regressions.

1. **Flat slicing.** For Minkowski space with $N = 1, N^i = 0, h_{ij} = \delta_{ij}$, all curvatures, momenta, and vacuum constraints vanish.
2. **Flat FLRW.** For $h_{ij} = a^2(t) \delta_{ij}$, $K_{ij} = H h_{ij}$. Equation (13) becomes $6H^2 - 2\Lambda = 2\kappa\rho$, hence $3H^2 = \kappa\rho + \Lambda$.
3. **Scalar momentum.** For the scalar model, $T_{ni} = (n^\mu \partial_\mu \phi) D_i \phi$, so $j_i = -p_\phi D_i \phi / \sqrt{h}$ and $-\sqrt{h} j_i = p_\phi D_i \phi = \mathcal{H}_i^\phi$.

B. Imported theorem-use ledger

The local reference in the third column is the exact theorem title, compiled number, or source label in the final source publication. The last column is part of the import: it prevents an accepted theorem from being promoted beyond its hypotheses.

Table 6: Hypothesis-level import audit.

Source	Hypotheses instantiated here	Conclusion used	Not imported
Kernel	Fixed domain, target determinacy, step evaluability, act-time finality, same-regime fidelity	<code>thm:main-fixed-domain-aasc</code> ; <code>thm:scope-preserving-invariance</code> ; <code>thm:transport-closure</code>	Metric or canonical structure
Unus	Instantiated constructional unit and fixed target	<code>thm:no-repairs</code> , <code>thm:no-generators</code> , <code>lem:no-scope-transport</code> , <code>thm:unique</code>	Unique foliation or source
RPI/AER	Regular local symplectic carrier, smooth standing surface, quotient observables, AER-Max	Local AER forcing theorem and gravity-role necessity	Infinite-dimensional global ADM quotient

Source	Hypotheses instantiated here	Conclusion used	Not imported
Einstein projection	Four-dimensional Lorentzian minimal first-response metric branch, closed source, fixed calibration	Einstein minimal projection and no metric backflow	Canonical phase space or quantum dynamics
Module A	Attachment certificate, complete continuous witness, naturality, smooth density certificate, closed Ward coordinate	Theorems 5.1, 6.4–6.5, 7.2–7.5, Example 13.2, Proposition 13.3, Ward closure	Cauchy trace of arbitrary distributions; matter Hamiltonian
Covariant Differentiation	Standard finite linear tangent/tensor carriers; smooth finite transport; minimal natural metric branch	Theorems 11.1, 17.2, 19.1	Superspace derivative or WDW operator
Non-Quantizability	Fixed common architecture and attempted same-domain foundational map	Theorems 2.10 and 5.3	Denial of effective or future certified quantization
Physical Clock	Closed constrained target, undeclared external parameter, fixed clock comparison class	<code>thm:no-external-time</code> ; lapse/shift boundary guard	No physical time in richer scopes
Arrow/Time	Standing-bearing irreversible continuation and downstream temporal use	No primitive temporal parameter; local partial temporal order	Lorentzian geometry or ADM split
Unification capstone	Frozen FEP-1.0 dependency and accepted Modules A–D	Theorems 10.1, 11.1, 14.1, 21.1, 82.1	All interacting quantum gravity

C. Connection branch occupancy and exclusion ledger

The statement $\nabla = \nabla^{\text{LC}}$ is meaningful only with its comparison class attached. This appendix audits that class against common alternatives. The point is not to deny the alternatives; it is to prevent their additional data from being smuggled into the minimal branch and then used as putative counterexamples.

C.1 The seven occupancy coordinates

The certificate $\mathfrak{C}_{\nabla}$ can be expanded as follows.

1. **Carrier.** The connection acts on TM and its naturally induced finite-rank tensor bundles. No infinite-dimensional configuration-space carrier occurs.
2. **Selector data.** The metric g is the only connection-selecting field. Time orientation selects a connected cone, not connection coefficients.
3. **Locality.** The value of the connection at p depends on a germ of the selector data at p , not on a Green function, global average, or boundary solution.
4. **Naturality.** For every admitted diffeomorphism φ , $\nabla[\varphi^*g] = \varphi^*\nabla[g]$.

5. **Jet order.** Only j^1g is available. Curvature is second-jet data and is therefore outside the selector grammar.
6. **Comparison preservation.** The finite transport preserves the admitted metric comparison, giving $\nabla g = 0$.
7. **No extra authority.** A source, foliation, boundary normal, tetrad, independent one-form, torsion field, or optimization functional is not available to choose among connections.

At a point p , choose normal coordinates for g . The metric components equal the Minkowski matrix and their first derivatives vanish. A natural tensor $C^\rho_{\mu\nu}(j^1g)$ representing the difference from Levi–Civita must therefore be built at p from the metric and invariant index operations alone. No tensor of type $(1, 2)$ with the required connection-difference transformation and no auxiliary vector or form is available. Naturality transports the zero value to every coordinate system and point. This is the local content of the imported difference-tensor vanishing theorem.

C.2 Alternative branches

Table 7: Connection alternatives and the coordinate they change.

Alternative	Added or changed datum	WDW-1 classification
Metric-compatible torsion	contorsion tensor or spin/torsion source	Richer selector/field target; not a same-branch survivor.
Weyl connection	one-form controlling nonmetricity	Adds standing-bearing tensor data.
General metric-affine	independent connection and its equations	Changes carrier and variational target.
Palatini	connection varied independently of the metric	Changed action and constraint architecture.
Teleparallel	tetrad and flat torsionful connection	Adds frame carrier and local Lorentz ledger.
Foliation-adapted	n^μ , K_{ij} , or preferred slicing selects transport	Violates the before-foliation ordering.
Source-adapted	$T_{\mu\nu}$ or matter current selects transport	Adds source-dependent connection data.
Curvature-selected	$R_{\mu\nu\rho\sigma}$ or its derivatives select ∇	Higher-jet branch.
Boundary-selected	boundary normal, charge, or falloff selects ∇	Boundary target update.
Nonlocal	Green operator, holonomy optimization, or global minimizer	Violates locality and changes continuation data.

Proposition C.1 (Comparison-class precision). *Theorem W1.1 is a uniqueness theorem within $\text{MinMetricNat}_1(M, g)$, not a theorem that every enriched gravitational model must use Levi–Civita transport.*

Proof. Every row in the alternative table changes at least one coordinate of $\mathfrak{C}_\nabla$. It therefore fails the antecedent of the vanishing theorem rather than supplying a second occupant with the same signature. Conversely, any connection that preserves all seven coordinates has zero difference tensor by Lemma 7.3. $\square$

D. Source quotient and Cauchy-split faithfulness audit

The Module-A quotient and the 3 + 1 split perform different identifications. Confusing them would either erase source data or grant foliation data upstream authority.

D.1 Operational quotient

For a structured source class $x = [(p, a_p, w_p)]_{\sim_{\text{src}}}$, write

$$q_g(x) = [T_x]_{\sim_{\text{law}}}.$$

Module-A probe faithfulness states, in one transported metric envelope,

$$q_g(x) = q_g(y) \iff W_x(\delta g) = W_y(\delta g) \text{ for every complete compact metric probe } \delta g,$$

modulo the frozen boundary equivalence. This is stronger than component equality in one chart and weaker than a claim that the upstream response quotient alone determines a source. The attachment certificate remains part of the structured domain.

D.2 Foliation-relative bundle isomorphism

Fix $\mathcal{F}$. The bundle map

$$\mathcal{P}_{\mathcal{F}} : S^2 T^* M \longrightarrow \mathbb{R} \oplus T^* \Sigma \oplus S^2 T^* \Sigma, \quad T \longmapsto (\rho, j, S),$$

is pointwise invertible by (9). Thus no tensor information is lost once the foliation and its normal are retained. But $\mathcal{P}_{\mathcal{F}}$ depends on $\mathcal{F}$: another normal generally gives another triple. The foliation-relative triple is therefore not an upstream replacement for T . Its invariant meaning is its reconstruction profile together with the selected presentation.

Proposition D.1 (Two-stage faithfulness). *On the smooth core, the composite*

$$\hat{Q}^{\text{src}} \xrightarrow{q_g} \text{Src}_g^{\text{sm}} / \sim_{\text{law}} \xrightarrow{\mathcal{P}_{\mathcal{F}}} (\rho, j, S; \mathcal{F}) / \sim_{\text{spatial}}$$

is faithful relative to the fixed foliation and lawful spatial redescription. It does not identify data across different foliations or construct a physical refoliation quotient. This is tensor-representation faithfulness; the full reconstruction map separately carries $(\hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}})$ as an identity fibre and does not infer those structured coordinates from the tensor image.

Proof. The first map is faithful on complete metric probes by Module A. The second is invertible when $\mathcal{F}$, n , and h are retained. Spatial coordinate changes transform the three factors naturally and leave the reconstructed tensor fixed. A change of foliation changes the second map itself, so comparison across such maps requires the WDW-3 refoliation theorem. The final structured-source clause follows from Definitions 22.2 and 22.3. $\square$

D.3 Support and boundary inheritance

If T vanishes on an open spacetime region, every projected component vanishes on the intersection of that region with a leaf. Conversely, if all three projected components vanish for every leaf in a foliation covering an open region, reconstruction gives $T = 0$ there. A boundary-supported distribution can be invisible to compact interior probes, but it is not zero in the complete source object. It must be retained as $(T_{\text{bulk}}, T_{\partial}, \mathcal{B})$. This is why the source trace certificate and boundary certificate share a status coordinate.

D.4 Closed, open, and anomalous ledgers

The split also preserves the Module-A current classification:

$$J^\nu = 0 \iff \varepsilon = 0, q_i = 0$$

for every leaf. An open source has named nonzero (ε, q_i) and an external exchange carrier. A boundary flux is not automatically a bulk anomaly; it can be the boundary part of an exactly closed total ledger. WDW-1 uses only the closed bulk core, while preserving the richer statuses in $\mathfrak{E}_\Sigma$ and $\mathfrak{E}_\partial$.

E. Inverse-map and solution-class audit

This appendix expands the two directions of Theorem W1.5 so that its “prequotient” qualifier cannot conceal a missing equation.

E.1 Covariant to canonical

Fix $\hat{x} \in \hat{Q}_{\text{sm},0}^{\text{src}}$. Given

$$(g, \hat{x}, T_{\hat{x}}, \text{Att}_{\hat{x}}, W_{\hat{x}}; \mathcal{F}) \in \text{Sol}_{\text{cov}}(\hat{x}; \mathcal{F}) :$$

1. the temporal function and metric determine $N > 0$, N^i , and h_{ij} ;
2. ∇^{LC} and n^μ determine K_{ij} ;
3. the Legendre map determines π^{ij} ;
4. the source tensor determines $(\rho_{\hat{x}}, j_{\hat{x}}, S_{\hat{x}})$, while the split map carries $(\hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}})$ unchanged;
5. the nn and ni projections give the four constraints;
6. the ij projection gives the six evolution equations; and
7. the Ward identity gives the four source-balance components, only four of which are independent as a spacetime vector equation.

The source balance is not an eleventh Einstein equation. It is the compatibility identity required by Bianchi closure and, for a canonical matter occupant, by matter evolution.

E.2 Canonical to covariant

Given an ADM history in $\text{Sol}_{\text{ADM}}^{\text{preq}}(\hat{x})$, define g by (1). Its determinant is

$$\det g = -N^2 \det h < 0,$$

so it is a nondegenerate Lorentzian metric with the selected time orientation. Define K from π by (18). The metric Hamilton equation ensures that this tensor is also the geometric extrinsic curvature of the reconstructed metric. Define T' by (9). The history’s source-ancestry condition is the equality

$$T' = T_{\hat{x}},$$

not merely membership in an unspecified tensor image. The inverse map returns the retained tuple $(\hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}})$; it does not infer that tuple from T' .

Now set $E = G + \Lambda g - \kappa T_{\hat{x}}$. The scalar constraint says $E_{nn} = 0$, the vector constraint says $E_{ni} = 0$, and the momentum evolution says $E_{ij} = 0$. Projection completeness gives $E = 0$. The Ward split gives the covariant source ledger. Thus the output lies in $\text{Sol}_{\text{cov}}(\hat{x}; \mathcal{F})$.

E.3 Compositions

Applying reconstruction after splitting returns the original structured object:

$$\mathcal{R}^{\text{att}} \circ \mathcal{S}_{\mathcal{F}}^{\text{att}}(g, \hat{x}, T_{\hat{x}}, \text{Att}_{\hat{x}}, W_{\hat{x}}; \mathcal{F}) = (g, \hat{x}, T_{\hat{x}}, \text{Att}_{\hat{x}}, W_{\hat{x}}; \mathcal{F}).$$

Applying the split after reconstruction returns the original ADM history provided the metric Hamilton equation is among the history conditions:

$$\mathcal{S}_{\mathcal{F}}^{\text{att}} \circ \mathcal{R}^{\text{att}}(h, \pi; N, N^i; \hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}}; \rho_{\hat{x}}, j_{\hat{x}}, S_{\hat{x}}) = (h, \pi; N, N^i; \hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}}; \rho_{\hat{x}}, j_{\hat{x}}, S_{\hat{x}}).$$

Without the metric Hamilton equation, an arbitrary π need not equal the extrinsic curvature of the reconstructed metric. Without the spatial momentum evolution, the six spatial Einstein equations are missing. Without source reconstruction and Ward matching, the right-hand side of the covariant equation is not certified. These are genuine inverse-map conditions, not presentational conveniences.

The identities hold fibrewise before taking the disjoint union. Since both maps have source-base component $\text{id}_{\hat{Q}_{\text{sm},0}^{\text{src}}}$, the union cannot collapse two structured source classes by forgetting attachment or witness data.

E.4 Equivalence relation actually used

The inverse statement is taken modulo coordinate changes and foliation-preserving redescrptions that reconstruct the same $(g, \hat{x}, T_{\hat{x}}, \text{Att}_{\hat{x}}, W_{\hat{x}}; \mathcal{F})$. It does not quotient:

1. different Cauchy foliations of one spacetime;
2. large spatial diffeomorphisms with potentially distinct global or boundary action;
3. internal gauge transformations of a future matter sector;
4. transformations carrying nonzero boundary charges; or
5. singular histories lying outside the regular canonical chart.

This exact list is passed to WDW-2 and WDW-3.

F. Boundary-generator variation audit

For a smearing pair $\xi = (N, N^i)$, the variation of a bulk generator has the form

$$\delta H_{\text{bulk}}[\xi] = \int_{\Sigma} \left(A_{\xi}^{ij} \delta h_{ij} + B_{\xi ij} \delta \pi^{ij} \right) d^3x + \int_{\partial\Sigma} \Theta_{\xi}(\delta h, \delta \pi).$$

On the compact core the last integral is absent. On a boundary branch, a differentiable generator exists only if the allowed variations make the surface one-form vanish or if there is a surface functional $H_{\partial}[\xi]$ such that

$$\delta H_{\partial}[\xi] = - \int_{\partial\Sigma} \Theta_{\xi}(\delta h, \delta \pi).$$

The improved generator is

$$H_{\text{imp}}[\xi] = H_{\text{bulk}}[\xi] + H_{\partial}[\xi].$$

Definition F.1 (Proper and improper boundary transformation). Relative to a fixed boundary phase space, a transformation is *proper* when its differentiable generator vanishes on the constraint surface and carries no charge. It is *improper* or a boundary symmetry when the improved generator has a nonzero surface charge on that surface.

This definition is branch-relative. A smearing that is proper under compact-support boundary conditions may become an asymptotic translation with nonzero ADM energy under asymptotically flat falloff. Therefore “lapse is gauge” is not a target-independent sentence.

F.1 Temporal parameter classification

Suppose the allowed asymptotic lapse approaches a constant N_∞ and

$$H_{\text{imp}}[N]|_{\mathcal{C}} = N_\infty E_{\text{ADM}}.$$

If $E_{\text{ADM}} \neq 0$, changing N_∞ changes the physical generator. The associated asymptotic time translation is standing-bearing boundary structure. By contrast, in the compact core $H_\partial = 0$ and the bulk Hamiltonian vanishes on the constraint surface; the coordinate label remains presentation data unless a separate internal clock certificate is supplied.

F.2 Source flux

For a source current, Stokes' theorem relates bulk Ward balance to boundary flux. Schematically,

$$\int_{\mathcal{U}} \nabla_\mu T^{\mu\nu} X_\nu dV = \int_{\partial\mathcal{U}} T^{\mu\nu} X_\nu s_\mu dS - \int_{\mathcal{U}} T^{\mu\nu} \nabla_\mu X_\nu dV.$$

A nonzero boundary term may encode real exchange even when the combined bulk-plus-boundary ledger is closed. Omitting it changes continuation and therefore fails $\mathfrak{C}_\partial$.

Proposition F.2 (Boundary standing test). *If varying a boundary value of N , N^i , or matter data changes a differentiable charge, flux, or continuation, that datum is standing-bearing in the boundary target and cannot be removed by the proper bulk quotient.*

Proof. The variation changes the value or Hamiltonian vector field of the improved generator, or changes the source balance. It therefore changes admissible continuation or a readout on the fixed boundary phase space. By the anchor–tensor–skin ledger it is not skin. The proper bulk quotient removes only zero-charge transformations and cannot identify distinct values of the nonzero generator. $\square$

G. Full 3 + 1 calculation

G.1 Derivative split

For any scalar f ,

$$\nabla_\mu f = D_\mu f - n_\mu \mathcal{L}_n f.$$

For a spatial covector $v_\mu n^\mu = 0$, define the projected covariant normal derivative $\dot{v}_{\langle\nu\rangle} = h_\nu^\alpha n^\beta \nabla_\beta v_\alpha$. Then

$$\nabla_\mu v_\nu = D_\mu v_\nu - n_\mu \dot{v}_{\langle\nu\rangle} + n_\nu K_\mu^\alpha v_\alpha - n_\mu n_\nu a^\alpha v_\alpha,$$

while the projected Lie derivative obeys

$$h_\nu^\alpha (\mathcal{L}_n v)_\alpha = \dot{v}_{\langle\nu\rangle} + K_\nu^\alpha v_\alpha.$$

The basic normal derivative is

$$\nabla_\mu n_\nu = K_{\mu\nu} - n_\mu a_\nu.$$

G.2 Gauss contraction

Contract (3) with $h^{ac} h^{bd}$:

$$h^{\alpha\gamma} h^{\beta\delta} R_{\alpha\beta\gamma\delta} = {}^{(3)}R + K^2 - K_{ij} K^{ij}.$$

Since

$$2G_{nn} = h^{\alpha\gamma} h^{\beta\delta} R_{\alpha\beta\gamma\delta},$$

equation (5) follows. The mixed contraction of Codazzi gives

$$G_{ni} = R_{ni} = D_j K^j_i - D_i K.$$

These two contractions are the four equations containing no second normal derivative of h_{ij} ; they are therefore constraint equations on each leaf.

G.3 Scalar curvature and boundary divergence

The twice-normal Ricci projection gives

$$R_{nn} = -\mathcal{L}_n K - K_{ij} K^{ij} + D_i a^i + a_i a^i$$

in the frozen convention. Combining it with the Gauss contraction yields

$${}^{(4)}R = {}^{(3)}R + K_{ij} K^{ij} - K^2 + 2\nabla_\mu (n^\mu K - a^\mu).$$

Integration over a compact slab converts the last term into endpoint terms. The GHY completion cancels the normal-derivative variation when the induced endpoint metric is fixed. If a timelike or null boundary is added, this cancellation statement must be recomputed for that branch.

G.4 Legendre calculation

From

$$\delta K_{ab} / \delta \dot{h}_{ij} = \frac{1}{2N} \delta^i_{(a} \delta^j_{b)}$$

and

$$\frac{\partial (K_{ab} K^{ab} - K^2)}{\partial K_{ij}} = 2(K^{ij} - K h^{ij}),$$

one obtains (17). Then

$$\begin{aligned} \pi^{ij} \dot{h}_{ij} &= 2N \pi^{ij} K_{ij} + \pi^{ij} \mathcal{L}_{\vec{N}} h_{ij} \\ &= \frac{N\sqrt{h}}{\kappa} (K_{ij} K^{ij} - K^2) - 2N^i D_j \pi^j_i \end{aligned}$$

after integration by parts on compact Σ . Subtracting $\mathcal{L}_g$ gives (21)–(22).

G.5 Source Ward split

The normal projection can be reproduced without component coordinates:

$$\begin{aligned} -n_\nu \nabla_\mu T^{\mu\nu} &= -\nabla_\mu (T^{\mu\nu} n_\nu) + T^{\mu\nu} \nabla_\mu n_\nu \\ &= \mathcal{L}_n \rho + D_i j^i + K \rho + K_{ij} S^{ij} + 2a_i j^i. \end{aligned}$$

The spatial projection similarly gives

$$h_{i\nu} \nabla_\mu T^{\mu\nu} = (\mathcal{L}_n j)_i + K j_i + D_j S^j_i + \rho a_i + S_i^j a_j.$$

These calculations verify (10)–(11) without importing a matter action.

H. Source trace and regularity ledger

Table 8: Cauchy source branches.

Branch	Required certificate	WDW-1 disposition
Smooth	Smooth Module-A density representative; smooth foliation	Core theorem; ordinary restriction and contraction.
Sobolev	Trace order above the codimension-one threshold; products and curvature controlled in one declared scale	Possible target update; not silently inferred.

Branch	Required certificate	WDW-1 disposition
Radon measure	Disintegration or trace relative to the foliation; controlled contractions	Distributional extension only.
General distribution	Pullback theorem, e.g. $WF(T) \cap N^*\Sigma_t = \emptyset$; t -continuity in a declared topology	Excluded from core unless proved.
Surface layer on Σ	Explicit jump/shell formalism and side-dependent geometry	Changed source target.
Boundary-supported source	Boundary test space, flux, and surface tensor	Retained in $\mathfrak{C}_\partial$, never erased as improvement.

For the microlocal branch, the minimum audit is:

1. identify the conormal bundle of every leaf;
2. verify the wavefront exclusion or cite an alternative trace theorem;
3. prove contractions with n, h are defined;
4. specify the topology in which projected distributions vary with t ;
5. formulate the Ward split distributionally; and
6. retain boundary and junction terms.

Failure of any item narrows the theorem to the smooth core; it does not invalidate Module A’s covariant distributional representation.

I. Canonical-lift certificate template

Table 9: Machine-auditable $\mathfrak{C}_{\text{src}}^+$ template.

Coordinate	Evidence required	Failure status
$\mathcal{Q}_m, \mathcal{P}_m(\mathfrak{M})$	Fields, bundle types, canonical momenta, model-level topology/domain	No canonical lift
Ω_m	Closed two-form and nondegeneracy/presymplectic status	No Hamiltonian vector field claim
L_m or equivalent	Explicit action or canonical construction	No source-to-Hamiltonian promotion
$\mathbb{F}L_m$	Formula, rank, primary constraints, singular strata	Restrict or stratify
$\mathcal{P}_{\text{cpl}}(\mathfrak{M}), \chi_{\mathfrak{M}}$	Coupled model carrier and explicit selected-variable comparison map	No split handoff
SplitMatch	Pullback equality for the two-form; additive normal and spatial constraints, or exact named correction	No direct-sum or additive-constraint claim
$\mathcal{H}_\perp^m, \mathcal{H}_i^m$	Derived densities and smearing domain	No total constraints

Coordinate	Evidence required	Failure status
$\mathcal{D}_{\mathfrak{M}}$	All primary/secondary constraints; first-class, second-class, reducible, or downstream-unresolved status; multipliers, bracket domain, boundary improvements, and matter-specific second-class resolution	Branch ineligible
$\text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}), \text{Src}_{\mathfrak{M}}$	Complete-history domain and a presentation-invariant map to the structured Module-A source quotient	No source-faithful lift
FibreRet	Exact preimage over each class; attachment, witness, tensor, and every lawful pointed lift retained	Model/source mismatch
EMTMatch	Historywise equality to the tensor representative of $\text{Src}_{\mathfrak{M}}(\gamma)$ under metric variation	Symbol match insufficient
WardImp	Canonical equations imply or reproduce the same J^ν ledger	Continuation mismatch
BoundaryMatch	Action and Hamiltonian surface terms, charges, flux	Restrict to compact core
Reg_m	One common function-space/domain ledger	Formal-only branch

J. Boundary differentiability ledger

Table 10: Boundary certificate branches.

Branch	Allowed variations	Surface completion	Charge/flux	Closure here
Compact $\partial\Sigma = \emptyset$	smooth bulk; fixed endpoint induced metric	GHY at temporal endpoints	no spatial charge/flux	Proved
Finite timelike	Dirichlet, Neumann, or mixed boundary data	GHY/corners plus Hamiltonian term	quasilocal charge; source flux	Conditional
Asymptotically flat	falloff and parity	Regge–Teitelboim completion	ADM/Poincaré charges	Conditional
AdS-like	conformal boundary data	counterterm or equivalent prescription	boundary stress/charges	Conditional
Null	null metric data and corner conditions	null boundary action	null charges/flux	Not integrated

The word “conditional” means that Theorem W1.7 applies once the row’s certificate is actually supplied. It is not a claim that WDW-1 has constructed every listed formalism.

K. Field-theoretic AER handoff

Certificate coordinate	Owner	Required endpoint
$\mathcal{P}^s, \Omega^s, \text{Domain}$	WDW-2 foundation	Declared completion and well-defined (pre)symplectic form
Spatial constraint map and rank	WDW-2	Regular or explicitly stratified constraint surface
Spatial/internal quotient	WDW-2	Proper gauge equals declared quotient kernel; boundary charges retained
Observable separation/maximality	WDW-2 with WDW-3 compatibility	No hidden presymplectic remainder
Normal deformation family	WDW-3	Defined Hamiltonian vector fields on common domain
Deformation algebra	WDW-3	Closure with structure functions and boundary terms
Characteristic identification	WDW-3	Refoliation equivalence proved, not stipulated
Common boundary ledger	WDW-2/3 integration	Improper symmetries and charges preserved

L. Theorem dependency graph

G_0 : target and import consistency,
 G_∇ : connection branch occupancy,
 G_Σ : Cauchy-source descent,
 G_{ADM} : prequotient canonical reconstruction,
 $G_{\text{src-can}}$: strong source-family lift with realization fibres,
 G_∂ : boundary differentiability and charge typing,
 $G_1 : G_0 \wedge G_\nabla \wedge G_\Sigma \wedge G_{\text{ADM}} \wedge G_\partial$ (WDW-1 core),
 G_2 : WDW-2 proper spatial/internal quotient,
 G_3 : WDW-3 refoliation/deformation closure.

The eligible canonical matter branch adds $G_{\text{src-can}}$ to G_1 ; an attached forced branch closes only its typed history output and is blocked by Elig_{23} . The admissible execution order is

$$G_0 \longrightarrow G_\nabla \longrightarrow G_\Sigma \longrightarrow G_{\text{ADM}} \longrightarrow G_1 \longrightarrow (G_2, G_3).$$

No downstream gate repairs an upstream failure. In particular, neither quotient closure nor quantization can retroactively supply an absent Cauchy trace, incorrect ADM sign, or missing source attachment.

M. Hamiltonian functional-variation audit

This appendix independently checks the functional derivatives used in Section 20. It is included because a correct final constraint formula does not by itself certify a consistent choice of canonical momentum, density weight, curvature sign, and lapse derivative.

M.1 Kinetic term

Write

$$Q(h, \pi) = \pi_{ab}\pi^{ab} - \frac{1}{2}\pi^2, \quad T[N] = \int_{\Sigma} \frac{2\kappa N}{\sqrt{h}} Q d^3x.$$

At fixed contravariant density π^{ij} ,

$$\delta Q = \left(2\pi^i{}_k \pi^{jk} - \pi \pi^{ij}\right) \delta h_{ij} + (2\pi_{ij} - \pi h_{ij}) \delta \pi^{ij},$$

and

$$\delta h^{-1/2} = -\frac{1}{2}h^{-1/2}h^{ij}\delta h_{ij}.$$

Therefore

$$\frac{\delta T[N]}{\delta \pi^{ij}} = \frac{4\kappa N}{\sqrt{h}} \left(\pi_{ij} - \frac{1}{2}\pi h_{ij} \right), \quad (37)$$

$$-\frac{\delta T[N]}{\delta h_{ij}} = -\frac{4\kappa N}{\sqrt{h}} \left(\pi^i{}_k \pi^{kj} - \frac{1}{2}\pi \pi^{ij} \right) + \frac{\kappa N}{\sqrt{h}} h^{ij} Q. \quad (38)$$

Equation (37) supplies the non-shift part of (27); (38) supplies the quadratic momentum lines of (28).

M.2 Curvature and cosmological terms

Let

$$V[N] = -\frac{1}{2\kappa} \int_{\Sigma} N \sqrt{h} ({}^{(3)}R - 2\Lambda) d^3x.$$

For compact Σ ,

$$\delta \int_{\Sigma} N \sqrt{h} ({}^{(3)}R) d^3x = \int_{\Sigma} \sqrt{h} \left[-N \left(({}^{(3)}R^{ij} - \frac{1}{2}h^{ij}({}^{(3)}R) \right) + D^i D^j N - h^{ij} D^2 N \right] \delta h_{ij} d^3x.$$

The sign in the Einstein-tensor term reflects variation with respect to h_{ij} , not h^{ij} . Consequently,

$$\begin{aligned} -\frac{\delta V[N]}{\delta h_{ij}} &= -\frac{N\sqrt{h}}{2\kappa} \left(({}^{(3)}R^{ij} - \frac{1}{2}h^{ij}({}^{(3)}R) + \Lambda h^{ij} \right) \\ &\quad + \frac{\sqrt{h}}{2\kappa} \left(D^i D^j N - h^{ij} D^2 N \right), \end{aligned} \quad (39)$$

which is the curvature, cosmological, and lapse-Hessian part of (28). A spatial boundary would leave the integration-by-parts term visible and would therefore require H_{∂} .

M.3 Shift generator

On compact Σ ,

$$\int_{\Sigma} N^i \mathcal{H}_i^g d^3x = \int_{\Sigma} \pi^{ij} \mathcal{L}_{\vec{N}} h_{ij} d^3x.$$

Its functional derivatives are

$$\frac{\delta H_{\vec{N}}}{\delta \pi^{ij}} = \mathcal{L}_{\vec{N}} h_{ij}, \quad -\frac{\delta H_{\vec{N}}}{\delta h_{ij}} = \mathcal{L}_{\vec{N}} \pi^{ij},$$

with π^{ij} treated as a density of weight one. This proves both shift terms in the Hamilton equations. It also shows why a surface term is necessary when N^i generates a nontrivial asymptotic symmetry.

M.4 Lapse and shift variations

Since N and N^i occur linearly after the Legendre transform,

$$\frac{\delta H_g}{\delta N} = \mathcal{H}_\perp^g, \quad \frac{\delta H_g}{\delta N^i} = \mathcal{H}_i^g.$$

On a certified matter lift, the corresponding derivatives of H_m give $\sqrt{h}\rho$ and $-\sqrt{h}j_i$. Without that lift, the same geometric equations are available as Einstein projections, but these functional derivatives have not been constructed. This is the precise location of the source/canonical gate.

N. Source-forcing trace and scalar variation audit

N.1 Transformation of the spatial Einstein source

Write

$$\Pi^{ij} = K^{ij} - Kh^{ij}, \quad \pi^{ij} = \frac{\sqrt{h}}{2\kappa} \Pi^{ij}.$$

The source-dependent part of (7) is

$$(\partial_t - \mathcal{L}_{\vec{N}})K_{ij}\Big|_{\text{src}} = N\kappa \left(S_{ij} - \frac{1}{2}h_{ij}(S - \rho) \right).$$

The metric kinematic equation has no separate source term, so raising indices and taking the trace at fixed instantaneous (h, K) give

$$(\partial_t - \mathcal{L}_{\vec{N}})K^{ij}\Big|_{\text{src}} = N\kappa \left(S^{ij} - \frac{1}{2}h^{ij}(S - \rho) \right), \quad (40)$$

$$(\partial_t - \mathcal{L}_{\vec{N}})K\Big|_{\text{src}} = \frac{N\kappa}{2}(-S + 3\rho). \quad (41)$$

Therefore

$$(\partial_t - \mathcal{L}_{\vec{N}})\Pi^{ij}\Big|_{\text{src}} = N\kappa(S^{ij} - \rho h^{ij})$$

and hence

$$\dot{\pi}^{ij}\Big|_{\text{src,raw}} = \frac{N\sqrt{h}}{2}(S^{ij} - \rho h^{ij}). \quad (42)$$

In converting the remaining geometric trace expression to (28), the vacuum conversion sets the gravitational Hamiltonian residual to zero. On the sourced shell, Theorem 20.1 instead gives $\mathcal{H}_\perp^g = -\sqrt{h}\rho$. The trace conversion therefore differs from the vacuum one by

$$-\frac{N}{2}h^{ij}\mathcal{H}_\perp^g = +\frac{N\sqrt{h}}{2}\rho h^{ij}.$$

Adding this to (42) cancels the intermediate energy-density term and proves

$$\mathfrak{F}_T^{ij} = \frac{N\sqrt{h}}{2}S^{ij}.$$

No matter phase-space variable was used.

N.2 Independent scalar Hamiltonian variation

For the Klein–Gordon model, the lapse-smearred matter Hamiltonian is

$$H_\phi[N] = \int_\Sigma N \left[\frac{p_\phi^2}{2\sqrt{h}} + \frac{\sqrt{h}}{2}h^{kl}D_k\phi D_l\phi + \frac{\sqrt{h}}{2}m^2\phi^2 \right] d^3x.$$

At fixed (ϕ, p_ϕ, N) ,

$$\begin{aligned} -\frac{\delta H_\phi[N]}{\delta h_{ij}} &= \frac{N\sqrt{h}}{2} \left[D^i \phi D^j \phi + h^{ij} \left(\frac{p_\phi^2}{2h} - \frac{1}{2} D_k \phi D^k \phi - \frac{1}{2} m^2 \phi^2 \right) \right] \\ &= \frac{N\sqrt{h}}{2} S_\phi^{ij}. \end{aligned} \quad (43)$$

The shift density $N^i p_\phi D_i \phi$ has no metric dependence in these selected variables, so it adds no spatial-stress term. Equations (29) and (43) are independent derivations of the same forcing history by history. On γ_ϕ , the source-realization equality $\text{Src}_\phi(\gamma_\phi) = \hat{x}_\phi$ identifies this tensor with the retained representative $\hat{T}_{x_\phi}$.

N.3 DeWitt quadratic form

Define the inverse DeWitt form on symmetric covariant tensors by

$$\mathcal{G}_{ijkl} = \frac{1}{2\sqrt{h}} (h_{ik}h_{jl} + h_{il}h_{jk} - h_{ij}h_{kl}).$$

Then

$$\frac{2\kappa}{\sqrt{h}} \left(\pi_{ij} \pi^{ij} - \frac{1}{2} \pi^2 \right) = 2\kappa \mathcal{G}_{ijkl} \pi^{ij} \pi^{kl}.$$

This is a classical finite-point tensor on each leaf. WDW-1 does not promote it to a configuration-space metric with a measure, connection, Laplacian, or operator ordering. Those are separate WDW-5 obligations.

O. Constraint propagation and Bianchi closure

The reconstruction theorem requires the constraints on each leaf. A Cauchy-evolution theorem usually asks a different question: if the constraints hold initially and the spatial evolution equations hold, do the constraints remain zero? The answer on the smooth Einstein branch is controlled by the contracted Bianchi identity and the source Ward ledger.

Set

$$E_{\mu\nu} = G_{\mu\nu} + \Lambda g_{\mu\nu} - \kappa T_{\mu\nu}.$$

For constant Λ, κ and a closed source,

$$\nabla^\mu E_{\mu\nu} = 0.$$

Decompose

$$\mathcal{C} = E_{\mu\nu} n^\mu n^\nu, \quad \mathcal{C}_i = E_{\mu\nu} n^\mu h_i^\nu, \quad \mathcal{E}_{ij} = E_{\mu\nu} h_i^\mu h_j^\nu.$$

The equations $\nabla^\mu E_{\mu\nu} = 0$, projected normally and spatially, form a first-order linear system for $(\mathcal{C}, \mathcal{C}_i)$ whose inhomogeneous terms are linear in $\mathcal{E}_{ij}$ and its spatial divergence. On the evolution shell $\mathcal{E}_{ij} = 0$, the system is homogeneous:

$$\mathcal{L}_n \begin{pmatrix} \mathcal{C} \\ \mathcal{C}_i \end{pmatrix} = \mathbb{A}^k(h, K, N) D_k \begin{pmatrix} \mathcal{C} \\ \mathcal{C}_i \end{pmatrix} + \mathbb{B}(h, K, N, DN) \begin{pmatrix} \mathcal{C} \\ \mathcal{C}_i \end{pmatrix}.$$

The coefficient matrices are fixed by the convention ledger. On a regular local Cauchy domain for which uniqueness holds, zero initial constraint data remain zero. This is the standard constraint-propagation mechanism.

Proposition O.1 (Propagation compatibility). *The WDW-1 source and evolution equations are compatible with constraint propagation: the contracted Bianchi identity and the exact Ward ledger produce a homogeneous propagation system for the constraint residuals on the spatial-evolution shell.*

Proof. The geometric Bianchi identity gives $\nabla^\mu(G_{\mu\nu} + \Lambda g_{\mu\nu}) = 0$. Theorem W1.2 gives $\nabla^\mu T_{\mu\nu} = 0$, so $\nabla^\mu E_{\mu\nu} = 0$. Substitute the normal/spatial decomposition of E and $\nabla_\mu n_\nu = K_{\mu\nu} - n_\mu a_\nu$. Every derivative of E becomes a normal derivative of $\mathcal{C}, \mathcal{C}_i$, a spatial derivative of those fields, or a term containing $\mathcal{E}_{ij}$. Setting the spatial projection to zero removes the latter inhomogeneity. Standard local uniqueness for the resulting regular linear system gives the claim. $\square$

Nonclaim O.2 (No global PDE promotion). Proposition O.1 is a compatibility and local propagation statement. It is not global existence, global uniqueness, geodesic completeness, singularity avoidance, or a theorem through changing constraint rank.

P. Detailed scalar source-family lift audit

The Klein–Gordon model is sufficiently simple to expose every coordinate of $\mathfrak{C}_{\text{src}}^+(\mathfrak{M}_\phi, \text{Src}_\phi)$ without introducing an internal gauge sector. The model carrier and the fixed source fibre remain distinct throughout.

P.1 ADM decomposition and Legendre map

The scalar configuration and momentum carriers are

$$\mathcal{Q}_\phi^\infty = C^\infty(\Sigma, \mathbb{R}), \quad \Pi_\phi^\infty = \Gamma^\infty(\text{Dens } \Sigma), \quad \mathcal{P}_\phi^\infty = \mathcal{Q}_\phi^\infty \times \Pi_\phi^\infty$$

with their standard smooth Fréchet topologies. The common scalar–gravity functional domain is $\mathcal{A}_{\text{reg}}^\phi$ from Section 25.

The scalar action density becomes

$$\mathcal{L}_\phi = N\sqrt{h} \left[\frac{1}{2N^2}(\dot{\phi} - N^i D_i \phi)^2 - \frac{1}{2} D_i \phi D^i \phi - \frac{1}{2} m^2 \phi^2 \right].$$

Hence

$$p_\phi = \frac{\partial \mathcal{L}_\phi}{\partial \dot{\phi}} = \frac{\sqrt{h}}{N}(\dot{\phi} - N^i D_i \phi), \quad \dot{\phi} = \frac{N}{\sqrt{h}} p_\phi + N^i D_i \phi.$$

The velocity Hessian is $\sqrt{h}/N > 0$, so the scalar Legendre map is regular on the core. The Legendre transform gives

$$H_\phi = \int_\Sigma \left(N\mathcal{H}_\perp^\phi + N^i \mathcal{H}_i^\phi \right) d^3x$$

with (31)–(32). Together with the gravitational inverse (18), this proves

$$\text{Im } \mathbb{F}L_{g\phi}^{\text{red}} = \mathcal{P}_{\text{cpl}}(\mathfrak{M}_\phi) \cong \mathcal{P}_g^\infty \times \mathcal{P}_\phi^\infty$$

through the explicit regrouping map

$$\chi_{\mathfrak{M}_\phi}(h, \pi; \phi, p_\phi) = ((h, \pi), (\phi, p_\phi)).$$

P.2 Hamilton equations

The canonical equations are

$$\dot{\phi} = \frac{N}{\sqrt{h}} p_\phi + N^i D_i \phi, \tag{44}$$

$$\dot{p}_\phi = D_i(N\sqrt{h} D^i \phi) + D_i(N^i p_\phi) - N\sqrt{h} m^2 \phi. \tag{45}$$

Using $p_\phi = \sqrt{h} \mathcal{L}_n \phi$, their conjunction is exactly

$$\square_g \phi - m^2 \phi = 0.$$

Thus canonical continuation and covariant matter continuation match.

P.3 History realization and the exact source fibre

For every complete coupled canonical solution history $\gamma \in \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}_\phi)$, the two Hamilton equations reconstruct ϕ_γ , while the retained ADM equations reconstruct g_γ . Module A, Proposition 13.3 applied to this covariant field supplies

$$\text{Src}_\phi(\gamma) = [\mathfrak{p}_{\phi_\gamma}, \text{Att}_{\phi_\gamma}, W_{\phi_\gamma}]_{\sim_{\text{src}}}, \quad q_g(\text{Src}_\phi(\gamma)) = T^\phi[\phi_\gamma].$$

The construction is constant on the foliation-preserving presentation class because it uses the reconstructed covariant pair (g_γ, ϕ_γ) . It is generally not constant on the full model carrier: histories with different scalar data can and normally do have different tensor representatives. This variation is precisely why $\mathcal{P}_\phi^\infty$ is indexed by $\mathfrak{M}_\phi$, not by $\hat{x}_\phi$.

For the frozen covariant scalar realization, direct splitting gives the named history γ_ϕ of Definition 25.2. Its reconstructed scalar is the same ϕ used in the Module-A response presentation. Attachment graph equality and unique tensor representation therefore prove

$$\text{Src}_\phi(\gamma_\phi) = \hat{x}_\phi, \quad \gamma_\phi \in \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}_\phi; \hat{x}_\phi).$$

This is the pointed occupation needed by W1.6. No selection of a canonical history from a bare tensor is used.

P.4 Stress tensor and constraint matching

The normal derivative is

$$\mathcal{L}_n \phi = \frac{p_\phi}{\sqrt{h}}.$$

Direct projection of $T_{\mu\nu}^\phi$ gives

$$\rho_\phi = \frac{1}{2} \left(\frac{p_\phi}{\sqrt{h}} \right)^2 + \frac{1}{2} D_i \phi D^i \phi + \frac{1}{2} m^2 \phi^2, \quad (46)$$

$$j_i^\phi = -\frac{p_\phi}{\sqrt{h}} D_i \phi, \quad (47)$$

$$S_{ij}^\phi = D_i \phi D_j \phi + h_{ij} \left[\frac{1}{2} \left(\frac{p_\phi}{\sqrt{h}} \right)^2 - \frac{1}{2} D_k \phi D^k \phi - \frac{1}{2} m^2 \phi^2 \right]. \quad (48)$$

It follows term by term that

$$\mathcal{H}_\perp^\phi = \sqrt{h} \rho_\phi, \quad \mathcal{H}_i^\phi = -\sqrt{h} j_i^\phi.$$

Metric variation of the action gives the same tensor; this is EMTMatch, not merely equality of the two displayed projections.

The independent Hamiltonian variation is (43):

$$-\frac{\delta H_\phi[N, N^i]}{\delta h_{ij}} = \frac{N\sqrt{h}}{2} S_\phi^{ij} = \mathfrak{F}_{T_\phi}^{ij}.$$

This equality verifies the spatial-stress coordinate without defining the covariant forcing from the Hamiltonian answer.

P.5 Coupled split and complete Dirac ledger

On the declared coupled Fréchet carrier, the total canonical one-form in the selected ADM variables is

$$\Theta_{\text{cpl}} = \int_\Sigma (\pi^{ij} \delta h_{ij} + p_\phi \delta \phi) d^3x,$$

so exterior differentiation gives

$$\Omega_{\text{cpl}} = \chi_{\mathfrak{M}_\phi}^*(\Omega_g \oplus \Omega_\phi).$$

The Legendre transform has no gravitational–scalar velocity cross-term, and its lapse and shift coefficients are exactly

$$\mathcal{H}_\perp^g + \mathcal{H}_\perp^\phi, \quad \mathcal{H}_i^g + \mathcal{H}_i^\phi.$$

The scalar Dirac ledger is:

Sector	Scalar status	Disposition
Primary scalar constraints	none	scalar Legendre Hessian $\sqrt{h}/N$ is invertible
Internal/Gauss constraints	none	no internal gauge carrier
Second-class constraints	none	no Dirac-bracket reduction required
Reducibility relations	none	no internal constraint complex
Gravitational primaries	P_N, P_i	downstream-unresolved with deformation complex
Deformation secondaries	total $\mathcal{H}_\perp, \mathcal{H}_i$	downstream-unresolved pending G_2/G_3
Spatial quotient	downstream-unresolved	G_2 , not preclaimed here
Boundary improvement	zero on compact core	reopen under $\mathfrak{C}_\partial$

Thus every matter-specific constraint coordinate is resolved before the scalar branch is declared eligible. In the extended multiplier presentation, P_N, P_i are gravitational primary constraints and the total $\mathcal{H}_\perp, \mathcal{H}_i$ are deformation secondaries; both entries carry the certificate’s admitted DOWNSTREAM-UNRESOLVED classification until G_2/G_3 establish their full generator and characteristic status. The reduced scalar model has $\mathcal{D}_{\text{can}, \mathfrak{M}_\phi} = \mathcal{P}_g^\infty \times \mathcal{P}_\phi^\infty$ and $\mathcal{K}_{\mathfrak{M}_\phi} = 0$. The pointed handoff additionally retains Src_ϕ , $\text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}_\phi; \hat{x}_\phi)$, and γ_ϕ . Thus the common gravitational deformation and quotient burdens remain explicitly downstream without leaving a certificate coordinate blank.

P.6 Ward and boundary matching

The identity

$$\nabla_\mu T_\phi^{\mu\nu} = (\Box_g \phi - m^2 \phi) \nabla^\nu \phi$$

shows that the Hamilton equations imply the closed Ward ledger. Conversely, on a branch where $\nabla^\nu \phi$ is not identically degenerate, the field equation is detected by that ledger together with the canonical construction. The implication from Hamilton equations to the Ward ledger is the coordinate required by WardImp; the limited converse just stated is not used in W1.6. Compact Σ removes the spatial surface term in (45); fixed endpoint scalar data complete the variational ledger. This verifies WardImp and BoundaryMatch for the example.

P.7 Why the example does not generalize automatically

For Yang–Mills fields the Legendre map is gauge-degenerate and Gauss constraints occur. For fermions, tetrad/spin structure and second-class or graded canonical issues enter. Higher derivative matter can enlarge phase space or invoke an Ostrogradsky analysis. Effective dissipative sources may have no closed action presentation. Each change adds a coordinate to the target and must be certified independently. The scalar proof therefore establishes inhabitation of the interface, not universal occupancy.

Q. Target-update register and acceptance matrix

Q.1 Target-update register

Table 11: Changes that require a new target statement.

Changed coordinate	New obligation	Why the core proof is insufficient
Noncompact Σ	falloff, functional domain, surface generator	Integrations by parts no longer have zero surface term.
Spatial or null boundary	$\mathfrak{C}_\partial$ with corner/flux data	Differentiability and standing of boundary transformations change.
Distributional source	trace or pullback theorem	Ordinary restriction used in W1.2 is unavailable.
Open source $J \neq 0$	external exchange carrier and balance	Closed Ward/Bianchi matching changes.
Torsion/nonmetricity	new connection target and field equations	MinMetricNat ₁ is no longer occupied.
Tetrad/spin matter	local Lorentz and spin-connection constraints	Metric-only scalar certificate is incomplete.
Internal gauge matter	Gauss sector, reducibility, boundary charge	$\mathcal{G}_a$ must be constructed and quotiented.
Singular Legendre branch	stratified Dirac analysis	Constant-rank regularity fails.
Topology change	new global carrier and continuation rules	The fixed global splitting may fail.
Quantum realization	regulator, ordering, domains, anomalies, inner product	No such data occur in G_1 .

Q.2 Gate acceptance matrix

Table 12: WDW-1 and downstream gate matrix.

Gate	Required evidence	Pass condition	Failure disposition
G_0	fixed tuple and import ledger	no silent target motion or import promotion	stop
G_∇	$\mathfrak{C}_\nabla$	all seven minimal-branch clauses verified	change or narrow target
G_Σ	$\mathfrak{C}_\Sigma(T)$	ρ, j, S and Ward split well defined	smooth subtype or no split
G_{ADM}	full 3 + 1, Legendre, evolution, reconstruction	covariant and ADM histories reconstruct locally	stop
$G_{\text{src-can}}$	$\mathfrak{C}_{\text{src}}^+(\mathfrak{M}, \text{SrcM})$ and $\mathfrak{L}^+(\hat{x})$	coupled/split symplectic equality, additive constraints, complete Dirac ledger, history-to-source realization, exact fibre, EMT, Ward, boundary, and domain matching	forced history only
G_∂	$\mathfrak{C}_\partial$	differentiable generator and complete charge/flux ledger	compact core only

Gate	Required evidence	Pass condition	Failure disposition
G_1	all core gates and branch typing	typed handoff family and eligibility map constructed	no ambiguous downstream use
G_2	WDW-2 proof	proper spatial/internal gauge equals declared quotient kernel	no quantum work
G_3	WDW-3 proof	deformation algebra and characteristic refoliation status close	no quantum work

Q.3 No-quantum leakage test

A manuscript fails the WDW-1 scope lock if it uses any of the following as a proved WDW-1 operation: $\delta/\delta h_{ij}$ acting on a wave functional; a Wheeler–DeWitt operator; a factor ordering; a regulated product of functional derivatives; a quantum constraint commutator; a physical Hilbert-space inner product; a group-averaging measure; a quantum clock; or a semiclassical Born–Oppenheimer time. These phrases may appear only in explicit nonclaims or downstream handoff statements.

R. Claim-status and terminology ledger

Projection. A role-preserving presentation of an upstream theorem object; not an ontological generation map between physical formalisms.

Reconstruction. Fibrewise recovery of $(g, \hat{x}, T_{\hat{x}}, \text{Att}_{\hat{x}}, W_{\hat{x}})$ from a complete regular source-attached ADM history, with the structured source coordinate retained rather than inferred from a bare tensor.

Prequotient. Before completing the physical equivalence relation across spatial gauge, internal gauge, and refoliation.

Physical quotient. A faithful descent retaining every standing-bearing observable, boundary charge, and continuation distinction.

Multiplier. A variable entering the action without its own velocity and enforcing a constraint in the declared presentation.

Gauge/redescription. A transformation proved to lie in the lawful physical quotient kernel; multiplier status alone is insufficient for the complete verdict.

Boundary symmetry. A transformation with nonzero differentiable boundary generator or charge; it is not proper bulk gauge.

Source attachment. Proof that an operational witness realizes the independently fixed upstream response graph.

Strong source-family canonical lift. Certificate constructing model-level matter phase-space and constraint data, proving the coupled-to-split symplectic and additive-constraint comparison, completing the Dirac ledger, and mapping complete canonical histories to their exact structured source classes.

Pointed lift. A certified model, source-realization map, complete history, and proof that the history lies in the exact fibre over the named structured source class.

Forced handoff. A source-attached reconstructive history with a gravitational canonical carrier and covariant source forcing, but no asserted matter moment map or downstream constraint-algebra

eligibility.

Connection ancestry. The theorem chain from finite standing-preserving transport to connection type and, on the minimal metric branch, Levi–Civita closure.

External time. A parameter imported with same-scope physical authority but absent from the declared state, boundary, background, and readout ledgers.

Relational clock. A subsystem with a clock-standing certificate whose correlations descend faithfully to the physical quotient.

S. Publication verification

S.1 Publication checklist

- ✓ Exact source lineage recorded.
- ✓ Imported theorem hypotheses instantiated and prohibited promotions stated.
- ✓ Minimal connection branch verified before foliation.
- ✓ Convention ledger consistent across geometry, action, and source signs.
- ✓ Gauss–Codazzi constraints checked independently.
- ✓ Core source lies in the smooth Cauchy-admissible subtype.
- ✓ Ward split recombines exactly.
- ✓ Gravitational Legendre transform and extended/reduced presentations distinguished.
- ✓ Constraints and evolution reconstruct all covariant components.
- ✓ $\mathcal{P}_g^\infty, \Omega_g, \Phi_g, \mathcal{C}_{\text{vac}}$, smearing spaces, and functional domains defined.
- ✓ Both reconstruction maps preserve $\hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}}$ fibrewise.
- ✓ Source forcing derived independently and scalar Hamiltonian variation matched.
- ✓ No matter Hamiltonian asserted without a populated source-family certificate.
- ✓ Strong lift proves the coupled split, retains a complete Dirac ledger, and constructs the canonical-history-to-source realization map.
- ✓ Scalar and coupled Fréchet carriers, the joint Legendre image, and $\chi_{\mathfrak{M}_\phi}$ are explicitly defined at model level.
- ✓ The scalar realization map is presentation-invariant, its exact fibre over $\hat{x}_\phi$ contains the named history γ_ϕ , and off-shell points are not promoted to the closed source base.
- ✓ Every conditional canonical constraint map has a named domain, codomain, and internal constraint carrier.
- ✓ Vacuum, canonical, and forced handoffs form a disjoint exhaustive source-index partition and carry a unique eligibility status.
- ✓ Compact-core differentiability proved; richer boundaries conditional.
- ✓ Finite/local AER scope stated; WDW-2/3 obligations not preclaimed.
- ✓ No quantum operator used.
- ✓ Abstract, main theorem, conclusion, and handoff report the same endpoint.
- ✓ Adversarial verification map checked for publication.

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